

PERCEPTIONS OF AFRICAN AMERICAN STUDENTS ABOUT THEIR SUCCESS
IN MATHEMATICS AT A HISTORICALLY BLACK
COLLEGE AND UNIVERSITY (HBCU)

A Dissertation

Submitted to

The School of Graduate and Professional Studies

Department of Educational Leadership

Tennessee State University

In Partial Fulfillment

of the Requirements for the Degree

Doctor of Education

Davon Waheed Williams

December 2020

Keywords: African-American STEM students, Math Perceptions, HBCU experiences,
Math Success, Student Success

COPYRIGHT

PERCEPTIONS OF AFRICAN AMERICAN STUDENTS ABOUT THEIR SUCCESS
IN MATHEMATICS AT A HISTORICALLY BLACK
COLLEGE AND UNIVERSITY (HBCU)

December 2020

Davon Waheed Williams

© 2020
All rights reserved

To the Graduate School:

We are submitting a dissertation by Davon Waheed Williams entitled, “Perceptions of African American Students about their Success in Mathematics at a Historically Black College and University (HBCU).” We recommend that it be accepted in partial fulfillment of the requirements for the degree, Doctor of Education in Educational Leadership with a concentration in K-12 Education.

Kirmanj Gundi, Ed.D.

Chairperson

Eleni Coukos Elder, Ed.D.

Committee Member

Jewell Winn, Ed.D.

Committee Member

Orville Bignall, Ph.D.

Committee Member

Accepted for the Graduate School:

Robbie K. Melton, Ph.D.

Dean of the Graduate School

DEDICATION

To accomplish such a great job, there must be support from an entire community, and with a heart full of gratitude, I would like to dedicate this dissertation to my entire family.

Firstly, I thank my wonderful and loving wife, Felicia, for her love and support throughout this entire process. She made it possible for me to attend classes, conduct ongoing research, and spend important and necessary time away from family to complete this dissertation study.

To my children, Davon, David, Diara, and Dasia, who allowed me to be away from them to study, research, and participate in activities that were needed for completing all necessary dissertation study requirements.

To my mother, Deborah Hamilton, who's continuous and unwavering support and commitment to this journey and me has reached far beyond anything I could have ever imagined. You are the sole greatest inspiration to all of my educational attainments.

To my uncle Brian Smith and aunt Claudine Smith and my cousins Revan Smith and Brittney Smith Clark who shared their home with me for several years while continuing my educational journey I appreciate greatly the sacrifices you made to make my stay as comfortable as possible.

To my other family members who I was not able to mention individually solely due to space, I thank you for all of your phone calls words of encouragement and support.

This accomplishment is part of yours as well for no individual can accomplish such a great feat without a mountain of support.

To my father and mother-in-law William and Delores Morris, I thank you for encouraging words and timely support whenever unexpected events occurred. I enjoyed your visits during the summer and at Christmas which brought a little of the Bahamas here to Nashville. This dissertation study is also in loving memory of my Grandfather and Grandmother Carlos Alexander Cox and Corene Irene Cox, who were tremendous sources of inspiration and guidance in my life. They were both strong educators whose life lessons continue to reside with me today. They inspired me to dream, believe, and know that any and everything is possible if you have the faith to believe and the stamina to persist. It would have given me great pride for them to witness this special occasion in my life. I miss them both, and with a grateful heart, express my gratitude for their part in making me the individual that I am today.

ACKNOWLEDGMENTS

There were many who gave their time, commitments, and expertise to help in completing a task of this magnitude and it is my desire that no single individual is omitted.

First and foremost I must thank God for giving me the strength and determination to complete this project. It is Him who has placed me in this position to accomplish such a great undertaking.

I would like to thank my chair Dr. Kirmanj Gundi who was invaluable in providing needed structure and direction in this difficult process.

I would also like to thank my committee members, Dr. Eleni Coukos Elder, Dr. Jewel Winn, and Dr. Orville Bignall whose words, knowledge, and expertise were critical in this process. They made this difficult process a lot more manageable.

PERCEPTIONS OF AFRICAN AMERICAN STUDENTS ABOUT THEIR SUCCESS
IN MATHEMATICS AT A HISTORICALLY BLACK
COLLEGE AND UNIVERSITY (HBCU)

ABSTRACT

DAVON WAHEED WILLIAMS. Perceptions of African American Students about their Success in Mathematics at a Historically Black College and University (HBCU) (under the direction of **DR KIRMANJ GUNDI**).

The purpose of this qualitative study using a phenomenological research analysis was to understand and examine perceptions of African American students about what constitutes success in mathematics from their perspective. A sample of ten participants was selected from a population of 51 STEM students who were currently enrolled in Pre-Calculus II or Calculus I and had passed College Algebra or Pre-Calculus I with an A or B grade. Participants responded to a series of open-ended questions from a questionnaire and filled out an initial demographic profile questionnaire. Analysis of the interview questions was conducted using an in-vivo coding approach which allowed the words of the participants to form the initial codes. In addition, Dedoose which is a qualitative software was used to help with sorting and organizing large amounts of interview data. Among the findings of the study, five major themes emerged: (a) Perceptions of Math Success, (b) Pre-College Experiences and Math Success, (c) Self-belief and Motivation connected to Math Success, (d) The HBCU Experience and Math Success, and (e) Women in STEM and Math Success. Further areas of research may include: (1) a longitudinal study with some of the major relevant themes to better understand the

experiences of successful African American STEM students, (2) a study conducted at a PWI to determine if African American STEM students within that community would have similar findings, and (3) a comparative study could be conducted to determine if non-STEM majors would share similar findings as found in this study with STEM majors.

TABLE OF CONTENTS

COMITTEE PAGE	iii
DEDICATION	iv
ACKNOWLEDGEMENT	vi
ABSTRACT	vii
CHAPTER I	1
INTRODUCTION	1
Statement of the Problem	3
Purpose of the Study	4
Significance of the Study	4
Theoretical Framework	5
Research Questions	7
Limitations of the Study	8
Definition of Terms	8
CHAPTER II	11
LITERATURE REVIEW	11
Overview	11
Introduction	11
HBCUs Impact of Student Success	13
Increasing Student Learning Outcomes in STEM Related Areas	18
Utilizing High Impact Practices for Student Success	19
Student Access to High Quality Education	22
Student Preparedness for College Level Math Courses	24
Student Ability and Self-Motivation	27
Extra and Co-curricular Activities	29
Faculty Instruction in Preparing Students	35
Chapter Summary	37
CHAPTER III	39
METHODOLOGY	39

Research Design	39
Setting, Population and Sample	40
Instrumentation	41
Procedures for Data Collection.....	41
Data Analysis.....	43
Chapter Summary	43
CHAPTER IV	45
RESULTS	45
Participant Summary One (Shelton)	50
Participant Summary Two (Toran)	51
Participant Summary Three (Christian S).....	53
Participant Summary Four (Tamera)	55
Participant Summary Five (Jalyn)	56
Participant Summary Six (Destiny)	58
Participant Summary Seven (Sean).....	60
Participant Summary Eight (Harmonie)	62
Participant Summary Nine (Christian B).....	64
Participant Summary Ten (William).....	65
Perceptions of Math Success	68
Maintaining academic excellence.....	68
Linking to future career success	69
Overcoming success obstacles.....	70
Success factors.....	70
Success strategies	71
Support systems.....	72
Technology Implementation.....	73
Pre-College Experience and Math Success.....	74
Early academic beginnings.....	74
Expectations of high school vs. university	75
Family expectations.....	76

Family Support	77
Extracurricular activities.....	78
Self-belief and Motivation connected to Math Success	80
Believing in one's ability	80
Future math success.....	81
Having balance with extracurricular activities	82
Math success motivations.....	82
Promoting a lasting legacy for African American Female STEM success	84
The HBCU Experience and Math Success	85
HBCU effect on math success	85
Help from professors from HBCU institutions.....	86
Importance of the HBCU connections to math success	87
Women in STEM and Math Success	88
Access to STEM for African American females	88
African American women in STEM as role models.....	89
Society's view of where women should be.....	90
CHAPTER V	92
SUMMARY, CONCLUSION AND RECOMMENDATIONS.....	92
Summary of the study	92
Limitations.....	97
Overview of the Results.....	97
Research Question One (Perception of Math Success).....	98
Research Question Two (Pre-College Experiences and Math Success)	99
Research Question Three (Self-belief and Motivation connected to Math Success)	101
Research Question Four (The HBCU Experience and Math Success)	102
Research Question five (Women in STEM and Math Success)	105
Conclusions and Implications	106
Recommendations for Future Research	108
REFERENCES	110

APPENDICES	128
APPENDIX A: Additional factors related to student Success.....	129
APPENDIX B: Permission to use Semi-Structured Interview Guide.....	132
APPENDIX C: Interview Guide	133
APPENDIX D: Demographic Questionnaire.....	135
APPENDIX E: Department Head Instructor Request.....	137
APPENDIX F: Department Head and Instructors Approvals.....	139
APPENDIX G: IRB Approval Letter and Amendment	142
APPENDIX H: Informed consent statement	144
APPENDIX I: CITI Completion Certificate	146
VITA.....	147

CHAPTER I

INTRODUCTION

Chapter I provides a brief overview of the need for African American students to be successful in mathematics at the post-secondary level. In addition, areas of focus to be addressed will include: Statement of the Problem, Purpose of the Study, Significance of the Study, Theoretical Framework, Research Questions, Limitations of the study, and Definition of Key Terms related to the study.

In 2016, 3.5 million students graduated from high school in the United States (NCES, 2016). For these students, options available for postsecondary aspirations consisted of several paths, (a) attendance at four or two year College or University, (b) attendance in a trade and or certificate program, (c) some form of military enrollment, (d) a Gap Year, in which the student refrains from any post-secondary involvement and (e) seek available job opportunities. For students choosing to follow an academic path by attending a post-secondary institution a flurry of uncertainty sometimes follow (Choy, 1999). For example, students may begin to realize the need for prior academic seriousness before this point and may question within themselves whether or not they are prepared for the daily academic rigor required for success within a university setting. In addition to uncertainties, many incoming students are ill-prepared to master some of the simplest of university courses inclusive of first-year level general education courses and introductory freshmen level mathematics courses (Anderson, 1990). According to several authors and academic think tanks, minorities are the least represented in the STEM fields

throughout many universities and have difficulties with connecting content in STEM fields to real-life applications (Powell, 1990; Anderson, 1990). Mathematics is one such area where many students struggle and in particular African American students (Powell, 1990; Anderson, 1990). This dilemma presents a concern for students, and faculty and a major concern for university professionals whose funding in many ways is related to students' success especially retention and graduation rates (Arnold, 1999). This is also a huge issue relative to many universities driving agenda which is to educate all students in a manner that provides them with sustained efforts to make a valuable contribution to society. For many institutions, the minority population inclusive of the African American groups tends to be the group needing to most support because they are the least prepared. Due to this large scope of this issue, it is central for all stakeholders to take action and lend support to provide needed resources and processes to ensure the success of all students and in particular those of African descent who are enrolled in first-year level university courses especially mathematics.

Over the years, research has highlighted the complexities of academic success, and noted authors such as Astin (2001, 1993a) and Tinto (1987, 1998) have shown academic success to be an all-encompassing process that relies on a myriad of contributing connected factors both on a social and academic level. Numerous studies have supported the notion of academic success being an interconnected process and have attributed academic success to many factors, see Table 1 in Appendix A by Zhang and Aasheim (2011). However, for this study seven areas of focus will be considered, (a) The Pre-College Experiences/Student Preparedness (b) Students' Self-Efficacy and

Motivation, (c) African American students involvement in STEM fields, (d) Parental Socio-economic status, (e) Programs and procedures within institutions, (f) Extra and Co-curricular Involvement, and (g) Faculty Instruction.

Statement of the Problem

Low passing rates among first-year university students in mathematics courses especially College Algebra and Pre-Calculus present a major concern and challenge for university stakeholders. Research supports this concern where African American students are usually lagging behind other ethnic groups when it comes to preparation for post-secondary success (Gutierrez, 2000). When comparing African American males and females, female students are usually less likely to take math classes beyond Calculus I or pursue a STEM-related major (Brew, 2002). This circumstance in many instances prevents African American female students from perusing STEM careers and usually contributes to the low representation of African American females in STEM-related fields. It is also a concern to STEM industries pursuing diversity in the work environment (Anderson, 1990).

Furthermore, when examining classes with only African American students enrolled, the failure rates for traditional class settings (classes with formal teaching instruction) are 55% higher than nontraditional class settings (classes combined with both formal instruction and technology implementation) (Freeman et al, 2014). If current trends continue, the retention and graduation rates at postsecondary institutions will continue to be a concern, workforce diversity will stagnate and many African American

minority students will find themselves in a continued struggle to achieve academic and career success.

Purpose of the Study

Through conversations with African American students in Science Technology Engineering and Mathematics (STEM) disciplines who have achieved success in mathematics, this study seeks to explore and determine perceptions these students have about the contributing factors which benefited their overall success in mathematics and strategies used to foster academic success. In addition, female African American students from these STEM disciplines will be interviewed to understand how perceptions and notions have tried to prevent them from having similar success as African American males in obtaining STEM degrees.

In addition, this study will allow the study participants to provide rich, non-numeric data, and allow qualitative information to be captured providing a naturalistic approach to involved subject matter and allow meaning to come from the study participants' perspectives and give voice to a minority group whose voice is seldom heard.

Significance of the Study

This study has significance for a variety of stakeholders inclusive of, University professionals, faculty, students, high school professionals, and parents. University professionals can use this study to identify gaps in best practices relating to African American STEM students and put additional measures in place to provide the most successful outcomes for students. It will also help with funding stability by ensuring the

success of the university's mission of educating all students and creating students with good academic backgrounds. Furthermore, this study will help students to recognize the success of others with similar backgrounds and provide a snapshot of how similar success can be achieved. Faculty can use this study to better understand the African American learner and implement strategies and techniques for instruction that will utilize their unique background with aligned academic content. High school professionals can use this study to ensure curriculums are optimal for the African American learner and continue to recognize the needed and added value of having strong involvement of parents and communities. Finally, parents can use this study for a greater understanding of the importance of continued support and nurturing beyond both the elementary and middle school environment while maintaining relevant learning and academic environments within the home.

Theoretical Framework

The theoretical framework lens for this dissertation study relies on the contributions of two frameworks, socio-cultural theory, and constructivist theory. These theories were chosen and relied on to work in conjunction with each other to provide a lens whereby we can address this study and develop a well thought out research design. The data collected through the interview process will be analyzed using this lens as a framework. From the perspective of sociocultural theory, learning is viewed as a cognitive process that takes into account the social interaction between the learner and the environment in which that learning occurs (Germain, 1991). Simply put, learning occurs when the social environment lends itself to that process. In addition to Germain

(1991), Tisdell (2000) describes the relationship between faculty, classroom, and school culture and its impact on student learning experiences as a broad scope of the sociocultural theory perspective. Communicated another way, Rogoff (1995), states “learning is the interaction between the environment and individuals.”

The Constructivist theory considers the process by which students learn and what they learn (Cobb, 1994). For instance, if African American students do not identify mathematical success as being consistent with their culture or social interactions they may not achieve the success in mathematics as commonly needed and, in many instances, demanded. This cultural and social interaction is in line with the Vygotskian approach perspective which suggests, the mental construct of a learner is linked to the social influences surrounding the individual. Hence, to understand the individual one must understand the social aspect of where the individual is coming from. This social interaction constructs an interwoven learning structure (Vygotsky, 1981).

Mathematics is viewed and practiced as an analytic approach to solving both simple and complex problems. However, the construction of theories and principles in mathematics plays a significant role in that process. The study of mathematics is a continuous and ongoing process that aims to provide students with the opportunity to interpret, organize, and construct meaning and solution to mathematical problems (Cobb & Yackel, 1995). Learning from the constructivist perspectives, ensures students’ success experiences of mathematics are built from conceived mathematical concepts (Von Glasserfeld, 1995). In addition, mathematical success is also constructed through scaffolding whereby students are rewarded incrementally to master content at their

current level of understanding. This constructivist perspective combined with a sociocultural perspective will provide deeper insight and meaning in the understanding of mathematical success amongst African American STEM Students. In line with both approaches, the focus of this work was on how students are able to use both perspectives as a mechanism to produce optimal learning and have measurable success at the undergraduate level in early mathematics courses at the postsecondary level. Understanding that knowledge acquisition and content mastery in mathematics can be influenced by both a socio-cultural and constructivist approach gives reliability to the usage of both frameworks as guided by those in favor of a coordinated perspective (Cobb, 1994; Cobb & Yackel, 1995).

Research Questions

To achieve the answers to the study, the following research questions were developed:

- 1) How do African American STEM students perceive success in mathematics at a Historically Black College and University (HBCU)?
- 2) How have experiences of African American STEM students (e.g., past and present) contributed to their success in mathematics?
- 3) To what extent being successful in mathematics contributed to overall self-efficacy and motivation of African American mathematics students?
- 4) How has attendance at an HBCU provided opportunities to meet the needs of African American students in STEM related discipline?

- 5) Is there a perception of African American STEM female students that gender plays a role in their involvement in STEM related disciplines and success in undergraduate entry-level mathematics courses?

Limitations of the Study

This study considers only STEM students enrolled at Blue University who have passed a freshmen level algebra or pre-calculus course with an A or B grade and are currently enrolled in an upper-level mathematics course beyond Pre-Calculus I. Due to the design of this study, the findings will not be generalized to other non-STEM majors. This research study is significant to the researcher because similar circumstances were encountered by the researcher being both first-generation and low income in addition to having math success in university-level mathematics courses and of African descent. The researcher has also worked with many African American students in mathematics and has thought about what constitutes success from their perception, and perspectives and what allows one group to be successful while the other group struggles. As a point of emphasis, the researcher will “bracket” as much as possible to relay the full and true lived experience of the participants. Bracketing allows the researcher to be unbiased in the assessment of the data collecting and interpreting process (Carpenter, 2007).

Definition of Terms

- *Academic success* - Based on the student's grade point average, class rank, and standardized test score results (Bean, 2005, p. 224;)

- *ACT* - The American College Testing (ACT) is a standardized test for school students to check on academic achievement and college admissions and possible college readiness (ACT Profile Report 2017).
- *Attrition* - the inability of a school to maintain its students (Arnold, 1999).
- *Bracketing* - The process of putting aside one's own biases in the investigation to allow the true lived experience of the study participant to be portrayed (Carpenter, 2007)
- *Coding* - Used to provide meaning from data and create themes in data analysis Creswell (2007, 2012a)
- *First-generation college students* – Students who are the first member of their immediate family to attend college or university (Chen, 2005).
- *Graduation rates* - The measure used to determine successful completion of post-secondary institutions (Arnold, 1999).
- *HBCU* - Section 322 of Title III of the Black College and University Act defined a historically black college and university (HBCU) as one that existed before 1964 with a mission of educating predominately African American students.
- *NAEP* - The National Assessment of Educational Progress (NAEP) is the largest nationally representative and continuing assessment of what America's students know and can do in various subject areas. Assessments are conducted periodically in mathematics, reading, science, writing, the arts, civics, economics, geography, and U.S. history (NCES, 2016).

- *Persistence* - A student's postsecondary education continuation behavior that leads to graduation (Arnold, 1999).
- *SAT* - The SAT Reasoning Test (formerly the Scholastic Aptitude Test or Scholastic Assessment Test) is a standardized test for college admissions in the United States. The SAT is owned, published, and developed by the College Board, a profit organization in the United States. It was formerly developed, published, and scored by the Service, which still administers the exam. The test is intended to assess a student's readiness for college (SAT profile report, 2017).

CHAPTER II

LITERATURE REVIEW

Overview

In a review of the literature on student experiences regarding success in mathematics at the post-secondary level the following themes were apparent: (a) Increasing Student Learning Outcomes, (b) Utilizing High Impact Practices (HIPS) for Student Success, (c) Student Access to Quality Education, (d) Student Preparedness for College-Level Mathematics Courses (e) Student Ability and Motivation, (f) Extra and Co-curricular Activities, and (g) Faculty Instruction in Preparing Students. This literature review will explore the research in these areas.

Introduction

Research has shown that those with post-secondary credentials over the course of a lifetime are able to make more money when compared with those who only have a high school credential (Redd, 2000). Therefore, it benefits one to seek out and complete a post-secondary education venture. However, for most students completing high school, there is no complete clarity as to what that entails. What most academic professionals would say is that students seeking post-secondary enrollment must be academically prepared to meet the demands of a post-secondary environment. In regard to this, many students believe they are capable of meeting the demands required to be successful in the post-secondary environment. However, in many instances once enrolled, students find it difficult to navigate the post-secondary environment process. This difficulty navigating

the post-secondary environment is the place where institutions can best lend a hand to fully support students and implement strategies and programs to ensure the most efficient ways for students to succeed and hence matriculate through typical post-secondary curriculums (Johnson, 1989)

In 2016 there were 3.5 million students who graduated from high school within the United States according to the National Center for Educational Statistics (NCES, 2016). Many of these students selected universities and majors they believed were beneficial to them in meeting their desired educational goals. Most of these students were provided with literature and brochures that showed degree completion could be attained within a four or at least a five year period, however, for many, degree completion takes a considerably longer period of time and usually extends to six or more years for an undergraduate bachelor's degree (Wenglinsky, 1998). Further providing frustration to the process is the need for students to be both financially and academically prepared for the demands necessary for postsecondary enrollment and success. Besides the financial support, many students selected for the institutions are not completely prepared to handle the rigor of a four-year university, much less the challenges and demands that come along with being a university student. Early in the academic path, many students begin to have problems with freshmen level mathematics courses such as College Algebra and or Pre-calculus which are the gateway courses for higher level mathematics courses. With this roadblock in place, some students began to realize possible deficiencies they have in this subject area and hence look for ways to mediate within the environment or overcome the deficiencies. For students choosing to remain in the class, the likelihood of success

becomes extremely challenging. However, there are some who are prepared for the academic rigor and do very well within the academic environment.

In this review, we will examine some of the research that has been identified as being impactful on determining whether a student fails or succeeds academically in the post-secondary environment. One study found that “Successful academic performance will be defined as academic achievement, engagements and educationally purposeful activities, satisfactory acquisition of desired knowledge skills and competencies, persistence and attainment of educational objectives” (Kuh, Kenzie, Schuh, & Whitt, 2005, p. 7).

Numerous studies have shown the connections to identified factors related to student success and successful academic performance (see Appendix A). Some major factors attributed to student success and academic successes were inclusive of personal factors, family factors, social factors, and environmental factors (McKenzie & Schweitzer, 2001). To note, in many of the studies researchers had the greatest impact on selecting how the studies were to be carried out and to what extent students would be allowed to interject their perspectives and views. The single purpose of the students was to test variables and hypotheses outline by the researcher. Additional problems with these studies were presented by the limited quantitative research methods seeking only to validate outcomes that related to a quantitative approach analysis.

HBCUs Impact on Student Success

Section 322 of Title III of the Black College and University Act defined a historically black college and university (HBCU) as one that existed before 1964 with a

mission of educating predominately African American students. In the United States, there are more than 100 HBCUs located predominately in the Southeastern United States, the Virgin Islands, and the District of Columbia. Of this number, 40 are public four-year institutions, 49 are four-year private institutions, and 11 are two-year public institutions (NCES, 2016). Many of the public HBCUs were started out of the second Morrill Act passed in August 1890 (Christy & Williamson, 1992). The Morrill Act made the way possible for land grant universities with southern and border states to gain access to federal funds to develop land grant colleges. Their initial usage was primarily for vocational training. During this period agricultural, mechanical, and technical institutions such as North Carolina A & T and Florida A & M were founded. By the late 1800, there were 1,111 college degrees awarded by public HBCUs to African American students.

In the 1896 decision, *Plessy v. Ferguson*, the two-tier system of higher education, based on the incentive structure in the Second Morrill Act became law. Consequently, HBCUs began to become institutions that served primarily as training facilities for teachers to work in the segregated public schools. The increases of black high schools in southern urban areas set in motion a supply-demand chain in which availability of teaching positions drew more black students into HBCUs to qualify themselves for teaching positions (Roebuck & Murty, 1993). The landmark decision in *Brown v. Board of Education* in 1955 and the legislation developed to implement it improved the financial strains of many HBCUs. Title III of the Higher Education Act of 1965, which was devoted to “Strengthening Developing Institutions” was interpreted as primarily referring to HBCUs. Due to this view, many HBCUs benefited greatly from the federal

funds provided under Title III. These funds were used for faculty and student exchanges, curriculum improvements, faculty improvement programs, visiting scholars' programs, student services, and administrative improvements.

HBCUs were practically the only institutions of higher learning open to African Americans in the US for nearly a century. Today, there is about 20 percent of African American students going to college who choose to attend HBCUs and these institutions are responsible for 22 percent of all bachelor's degrees granted to African American students (Roland & Greenstone 2007). HBCUs have produced several noted graduates including, W.E.B. Dubois (Wilberforce University), Ralph Ellison (Tuskegee University), Martin Luther King, Jr. (Morehouse College), Thurgood Marshall (Lincoln Memorial University), Ruth Simmons (Dillard University), and Oprah Winfrey (Tennessee State University) just to name a few. Among African Americans, 40% of all congressmen, 12.5% of CEOs, 50% of professors at non-HBCUs, 50% of lawyers, and 80% of judges are HBCU graduates (Roland & Greenstone 2007). In addition, HBCUs awarded 40% of all black STEM graduate degrees and noted the largest number of African Americans who enrolled in STEM doctoral degree programs are from the top 30 HBCU's. Elliott, Strenta, Adair, Matier, & Scott (1996) suggest that despite weak pre-college academics admissions, HBCUs produced 40% of black science and engineering degrees with only 20% of the total black undergraduate students' enrollment (Moore 2000). Elliott Et al. (2006), also noted that of the top 21 undergraduate producers of black doctoral graduates during the period 1986–2006, 17 were HBCUs (Moore 2000).

According to Merisotis ET al. (2006), a national survey indicated that student

engagement pertaining to both in class and on campus was higher for African American students attending HBCU institutions compared to African American students attending a non-HBCU institutions. Of importance, was the fact that higher interactions were found between African American students and professors at HBCU institutions leading to a greater degree of academic success. In the 2004 Campus Compact United States Commission on Civil Rights (USCCR, 2006 p. 27) membership survey, it was documented that HBCUs are more insistent compared with other institutions on having students participating in service-learning activities prior to graduating. In his research Merisotis Et al. (2006) also found that HBCUs provide needed support and guidance along with mentoring to students from both educationally and economically disadvantaged backgrounds to ensure their progress from admission to graduation from colleges and universities. Merisotis et al. 2006) further point out the success of such programs as HBCU UP, which provides opportunities to disadvantaged student populations. HBCU UP is a program funded by the National Science Foundation (NSF) at 14 HBCU institutions. According to data from the NSF of institutions with the program for at least five years, significant improvements in passing required math courses and other STEM courses were observed. Of the STEM graduates from these HBCUs, about 25 percent have acquired undergraduate research experience that will help them in graduate school if they choose to do graduate work (USCCR, 2006 p 31).

Four factors for educational success of African American students were listed by several noted researchers such as Astin (2001) and Pascarella and Terenzini (1991) which include: (a) faculty members' dedication to teaching, (b) HBCUs' supportive social

environment, (c) Strong encouragement to students to explore career options and leadership roles in their chosen careers; and (d) faculty role models (USCCR 2006 p 14)

Not to be outdone by other researchers Merisotis Et al., (2006) listed four areas of HBCU effectiveness which included, (a) HBCUs as leaders in student engagement, (b) HBCUs as community-based institutions that promote civic engagement and service-learning, (c) HBCUs as drivers of educational attainment for low income, first-generation, and disabled students, and (d) HBCUs as examples of success in a national effort to improve the quality of science, technology, engineering, and mathematics (STEM) in education and research (USCCR, 2006 p, 24).

Since their inception, HBCUs have been committed to the educational advancement of the disadvantaged population inclusive of low-income, first-generation, and disabled students. In doing so, these institutions work hard to provide minority students with additional support, guidance, and mentoring that will improve their opportunities to get into, and succeed in, college environments. One such program funded by the federal government which has been a long supporter of providing increased opportunities to such disadvantaged populations is the federal TRIO programs. This program contains a group of seven programs authorized under the Higher Education Act to provide a continuum of services from pre-college to pre-graduate level study for the nation's low-income, first-generation, and disabled students.

The full impact of the effectiveness of HBCUs on African American students is monumental however, one can begin to grasp a broad perspective on how effective HBCUs can become by seeing how resources are in place to ensure full support of

minority students who are often first-generation and low-income students. In spite of difficulties facing many of the nation's HBCUs, they tend to be very effective in attracting, retaining, and graduating a vast majority of African American students. Research has shown, African American students at HBCUs are very likely to complete bachelor's degrees and have better on-campus experiences (Pascarella & Terenzini, 1991). Data measures such as graduation rates and retention rates only show a small part of the larger picture. However, it is vitally important to remember the other benefits which include civic engagement, faculty interactions, supportive environment, faculty role models, leadership opportunities, and high success rates in STEM-related fields.

Increasing Student Learning Outcomes in STEM Related Areas

The failure of the academic pipeline has contributed to the inability of the United States to accomplish a diverse workforce in the area of science, technology, engineering, and mathematics (STEM). This has caused an unregulated and declining flow of underrepresented minorities inclusive of African Americans in the STEM field. Efforts to correct this issue have started but are not extensive enough to alleviate the problem.

According to the National Science Board (2010), there is data that lends support to the notion of declining interest of students wishing to pursue STEM majors' especially female minorities and those from low socioeconomic backgrounds. Johnson (1989), contends, even when students designated interest in STEM at the high school level and starts the STEM matriculation process at the postsecondary level the outcomes for success in STEM-related courses especially mathematics, are considerably low. Even though both African American men and women continue to leave STEM at a higher rate

than their white counterparts' the rates at which minority women leave the STEM discipline are much higher than both groups (Anderson & Kim, 2006).

According to the National Center for Educational Statistics who tracks persistence across all universities, it has been found that the most likely group to leave STEM majors are African American students who either drop out of college (29%) or those who decide to switch to another non-related STEM discipline (36%) (Anderson & Kim 2006).

A core component of STEM initiatives is to improve postsecondary outcomes by increasing the strength and rigor of Pre K-12 programs (Ackerman, Kanfer, and Calderwood 2013). According to Ackerman, Kanfer, and Calderwood (2013), there is documented evidence that supports the relationship between taking Advanced Placement (AP) courses with persistence in STEM-related outcomes. Also found were students who persisted highest in STEM-related majors had taken three or more AP exams while taking advanced level classes in high school.

Utilizing High Impact Practices for Student Success

Kuh et al. (2005) suggest, "For students to have the highest level of engagement and student success there should be opportunities for students to participate in at least two high impact activities during the college matriculation process". Kuh Et al. (2005) further insist these opportunities should take place during the freshman year but also during a later period where students can implement intellectual learning gathered from the experiences. Kuh Et al. (2005) List 10 High Impact practices as follows:

First-Year Seminars and Experience

These experiences afford students the opportunity to interact with both faculty and staff on regular basis and seek to increase critical thinking skills among students, create an environment for creative writing, and produce opportunities to increase student intellectual and practical competencies

Common Intellectual Experiences

Moving away from the idea of a common core systems the common intellectual experiences places an emphasis on a common set of core course which provides students a unique opportunity to become involved in advanced integrative studies and provides a broad theme to students learning experiences inclusive of curricular and co-curricular options

Learning Communities

Encourages students to solidify learning across multiple platforms and answer questions that have significant implications. In the learning community setting students are grouped with other students and professors via similar course interest and teaching interest. The goal is to explore common interests through the lenses of various disciplines.

Writing-Intensive Courses

The purpose of this unique experience is to allow students to write in a meaningful way over every aspect of the university matriculation period. Writings are geared toward both different audiences and disciplines. With mastery in this area, students have shown success in several ways, increased knowledge in quantitative reasoning, oral proficiency, literacy, and ethical Inquiry.

Collaborative Assignments and Projects

Two key components are afforded in this area, learning to solve problems in the company of others and gaining insight by listening to the perspective of others

Assignments may be enforced by study groups within a class or represented by joint efforts within a project setting

Undergraduate Research

These opportunities give students an early involvement in systematic investigation and research to help with the solutions of key concepts and questions.

Students are involved with empirical questions and current technologies

Diversity / Global Learning

Allow students opportunities to partake in different cultures and experiences that are unique from their own. In addition, students are given opportunities to explore difficult areas such as racial, ethnic, and gender inequalities from a global perspective

Service Learning, Community-Based Learning

Afford students the opportunity to work with community partners and align theoretical knowledge with practical knowledge. Students are afforded the opportunity to give back to the community while becoming good citizens and gaining experiences to help in both life and work opportunities.

Internships

These are experiential learning opportunities, which provide students with coaching and supervision from professionals in their related fields.

Capstone Course and Projects

Provides the student with an opportunity to create a project nearing the end of their college years that encompasses all the training acquired over the college matriculation

Student Access to High Quality Education

When economical resources are available, it is easier for families to provide rigorous academic preparation to their children and afford opportunities to set high educational aspirations. If we consider this from another viewpoint, as family income increases students in those environments are able to receive the necessary resources that place them on a path leading to greater success. In many instances, family socioeconomic status (SES) put in place the tools necessary for students to perform better academically in both the home and the school environment (Choy, 1999).

Students, whose families possess wealth, have more access and control to determine the kind of school and classroom environment they will be placed in (Reynolds & Walberg 1992). Also, families whose SES status is favorable tend to live in affluent areas and pay higher taxes which in turn help with school funding initiatives in the communities where they live (National Research Council, 1999).

In areas where property values are high and occupants of neighborhood housing are of a higher SES, schools tend to spend a lot more per student (Choy, 1999). Wenglinsky (1998) noted many of these schools had different levels of instruction, students had greater access to materials, teachers were better trained and adequately qualified to teach difficult subjects and the ratio of student to teacher was considerably

lower. Families whose socioeconomic status was higher had a better quality relationship with teachers and school personnel (Watkins, 1997). According to (Astin, 1993a) a student's socioeconomic status provided the best predictor of earning a bachelor's degree after taking into consideration academic ability.

According to (Choy, 1999) Students who completed high school and were from low-income families were 49 percent less likely to attend a two-year or four-year college or university compared to middle-income families, which accounted for 63 percent and high-income families, which accounted for 78 percent of college enrollment. In some ways, one can argue that students from low-income families are far worse off if they plan to attend college because they may go, and not be able to finish due to financial hardship. When this happens, debt is accrued and in many cases, students are not able to pay loans because they do not graduate and the low paying jobs acquired do not meet the financial obligation (Gladieux & Swail, 1998). Students, whose families had higher incomes, were more likely to pursue post-secondary education, and their interest in completing college applications and gaining admission to colleges and universities were higher. These were students who made sure to take advantage of advanced placement courses in high school. Low-income families accounted for 53% of students who were qualified for college compared to 68% of middle-income children and 86% of high-income children (Choy, 1999).

This trend is not limited to solely the African American race but seems to extend into different races as well, for example, middle-income Latinos had a 17% higher probability of earning a bachelor's degree compared with low-income Latinos (Swail,

Cabrera, Lee & Williams, 2005). In a similar fashion 46% of eight grades, Latino students tendered to have low-income backgrounds compared to white children which were at 17% (Swail et al., 2005). In many instances, the college selection of most students is determined by the ability of their parents to afford that school. When students from low-income schools are academically qualified and are able to take the necessary steps to enroll, there is still a lagging disadvantage to similar students enrolled in a high performing school 83% compared to 92% (Choy, 1999). Students whose family income exceeded \$60,000, were considered more likely to enroll in a post-secondary option compared to families with income between \$30,000 and \$59,999 (Choy, 1999).

Student Preparedness for College Level Math Courses

Prior to the 1900s, students' academic preparedness and financial aid procedures led to too much of the discussion in regards to access and educational attainment (St. John, 2003; Tierney, Corwin, and Colyar 2004; Tierney and Hagedorn, 2002; Wilkinson, 2005). Most recently, it has been shown through research that many students lack understanding about post-secondary options especially those who are coming from underserved and underrepresented backgrounds. These students are unaware of the expectations required for successful college work and attendance which include; academic workload, tuition and fees, and examinations required for college entrance such as the ACT and SAT (Venezia, Kirst, & Antonio, 2003).

There is a considerable amount of information available through research that show how postsecondary encouragement programs can address numerous concerns of low academic success students, and help improves access for students from traditionally

low and moderately low-income families, and in addition address the same issue with first-generation students (Hossler, Schmit, and Vesper 1999; Tierney, Corwin, and Colyar 2004; Tierney and Hagedorn 2002). Parental involvement and college outreach programs seem to be particularly effective and several programs have addressed this need, for example, Parent Institute for Quality Education and the Puente Project which have brought together minority families, students, teachers, and counselors to learn more about post-secondary options and financial aid (Chrispeels and Rivero, 2001; Pathways to College Network [PCN], 2004).

GEAR UP, which is one of the largest and best-funded initiatives, provides information about financial aid, family support, counseling, and tutoring, among other things (Hossler and Schmit 1995; St. John 2003; Tierney and Hagedorn, 2002). Also helping with the concern is the study of the 21st Century Scholars (TFCS) program which provides financial assistance and support to students from lower-income families to encourage participation in postsecondary education. Students in this program at times do not take full advantage of the benefits even though TFCS pays the tuition and fees at 4-year public institutions for students who complete the program. There are many students who qualify to enter this program but usually do not, and fewer than 60% of those who enroll after completing high school matriculate to post-secondary institutions (Tierney & Hagedorn, 2002).

Other promising encouragement initiatives include many of the TRIO programs funded under Title IV of the Higher Education Act such as Upward Bound, Upward Bound Math/Science, Student Support Services, Talent Search, Educational Opportunity

Center, and McNair Program (Institute for Higher Education Policy [IHEP], 1995, 2001; PCN, 2004). Students who participate in these programs are more likely to go to college, for example, students in the upward bound program are four times more likely to earn a bachelors degree compared to those not enrolled in the program. Another good statistic about the TRIO support program is that students who participate are twice as likely to remain in college compared to those who did not participate in the TRIO programs ([IHEP], 1995, 2001; PCN, 2004). Students in the TRIO program with the lowest educational aspiration who stay the longest tend to do better at the college level and those who remain in the Upward Bound program increased their chances to attend post-secondary education by nine percent each year (Meyers et al. 2004; Muraskin, 1997). Programs with a strong residential component are among the most effective, allowing students to become familiar with the physical, social, and cultural environments of the campus where they will matriculate (Barovick & Baron, 2001).

Credit-based programs offer greater incentives to high school students because they allow students to receive college-level credit for classes (Hughes, Karp, Fermin, & Bailey, 2005). There are approximately 57% of post-secondary institutions that have dual enrollment programs (Hoffman, 2005; Kleiner & Lewis, 2005). These programs reach out to students who are in need of academic support or to students who are interested in getting a head start in college. Another advantage suggested by Pennington (2004) is to provide accelerated course work through grade 11 which introduces competency-based models, allowing more choice and increases competition among high schools, private schools, community colleges, and universities competing for students; and expanding

learning options during the summer. Terenzini, Springer, Yager, Pascarella, & Nora, (1996) noted that such a transition program between high school and college would validate first-generation type students in helping them to successfully transition to university. Another set of pre-college programs that seems to work fairly well is the supplemental instructions programs which include some forms of tutoring services, these type of approaches can be used at home, in libraries, and also with other mentoring and tutoring programs (Gordon, 1999).

Student Ability and Self-Motivation

Students trusting their abilities in academic settings is a critical factor for academic success because it allows students to put their self-efficacy to the test (Hill, 2002). According to Gardner (1983) and Bandura (2001), self-efficacy is the ability of an individual to believe in their own capabilities regarding any course of action in which they try to reach an attainable goal. Self-efficacy can be classified into several categories, one being the academic ability of the student, which speaks to the students' ability to handle tasks related to academic endeavors (McCombs & Marzano 1990).

Cognitive abilities and academic self-efficacy have been recognized in the literature as well-established predictors of academic success (Dweck, 1999). However, the relationship between cognitive abilities and academic self-efficacy has not been sufficiently explored (Schunk, 2004). Social-cognitive theory declares that self-efficacy beliefs originate from a larger framework of "self-theories" (Dweck, 1986) that encompass motivation as well as performance. These theories cause people to develop beliefs about themselves that determine the process of thought they think as well as

develop the construct of the psychological worlds, which shape our feelings and behaviors. This approach gives reasons as to why some students are motivated to work harder, and why other students become helpless and are self-defeating. From this perspective, two implicit theories apply, (a) entity theory of intelligence, and (b) incremental theory. Entity theory asserts that intelligence to be fixed and stable and there is no chance of that changing and incremental theory entertains the idea that intelligence is malleable, fluid, and changeable (Dweck, 1999).

According to research, it has been found that students who believe in incremental intelligence display a higher level of self-efficacy are far more motivated, and are more able to diversify and creating a broader thinking style (Komarraju & Karau, 2005). When thinking about self-efficacy it can be shown that there exists some relationship with self-regulation (Wolfe & Johnson, 1995; Tuckman, 2003). However, there exist many inconsistencies when it comes to goals, achievement, and motivation. Students adjust their motivation and persistence to the task at hand in the way that is most comfortable for them. For example, when thinking about a task they may be required to do, performance strategies may consider looking at it from the perspective of performance goals (Komarraju & Karau, 2005). The relationship between self-efficacy and academic success is also connected to concerns such as self-regulation, which interjects implicit theories of intelligence and achievement goals in multiple and complicated avenues. Representing self-efficacy within the framework of social-cognitive theory, Bandura, (2001) describes self-efficacy as a “motivational orientation that fosters determination when faced with difficult situations, reinforces purposeful actions, inspires long term

goals, nurtures self-regulation and allows for individual correcting where ever there is a need for it.”

Self-efficacy has been analyzed and realized as a predictor of academic success and performance, which remains constant over time and is also stable in different environments and communities (De Raad & Shouwenburg, 1996; Duckworth, Peterson, Matthews, & Kelly, 2007; Farsides & Woodfield, 2003). In his studies, Abouserie (1995) supports the idea that strong or weak levels of self-efficacy are directly related to the success or failure of university students. The research literature also provides information as to how the motivational impact of self-efficacy gives birth to academic success (Ashwin, 2006; Pritchard & Wilson, 2003; Ridgell & Lounsbury, 2004; Chamorro-Premusic & Furnham, 2003). Miller & Brickman (2004) reported that confidence in a person’s ability is related to robust academic performance and may influence students to pursue and complete tasks that are difficult but within the scope of the students' ability. Furthermore, Frey & Determan (2004) assert that students who display exceptional abilities display better performance and receive higher grades. These students tend to display higher self-efficacy and lower anxiety. Another finding by Livengood (1992) asserts the same set of facts when looking at students within a single semester that continuously received feedback about their performance. Students who did well noted higher self-confidence and those who did poorly stated they were not as confident in their abilities.

Extra and Co-curricular Activities

Students involved in on-campus activities are generally considered to have an

advantage over those who do not participate in on-campus activities (Astin, 2001). When students for example participated in the on-campus activity of Western Kentucky University (WKU) after enrollment they received not only information's about the university but also received help with communication skills, professional development issues, and group gestures (Pascarella & Terenzini, 1991). Students who are satisfied with college life and are happy are usually the ones who continue to remain and excel in the collegiate environment (Lau, 2003). The more involved students become in their university the more likely they are also to matriculate and graduate (Astin, 2001).

In some instances, students can become too involved with on-campus life and lose sight of the true reason for being on the campus in the first place which is to excel academically and create successful academic outcomes (King, 2006). Research has displayed that students who are integrated into the university setting at an early stage are positively impacted and are more likely to persist and complete degree requirements (Pascarella & Terenzini, 1991). Pascarella and Terenzini (1991) discovered that students who connected with faculty, staff, and peers outside of the classroom environment structured educated opinions, attitudes, values, and aspirations based on those encounters. In addition, students involved in-class activities and are engaged can establish their identity and become self-governing (Astin, 2001). Astin's Student Involvement Theory suggests that when students are actively engaged there is a better outcome in their learning and academic environment (Hutley, 2004). Astin's Student Involvement Theory also supports the idea that when students are involved in all aspects of the college experience they learn more in all aspects. Astin's (2001) description of an involved

student is, a student who dedicates energy to academics, spends a great deal of time on campus, participates in student organizations and activities, and interacts with his or her faculty (Monroe Community College, 2004). Astin's Theory of Involvement is considered to be the most concise and sensible approach to student involvement and engagement, it is considered to be simple yet states the point that a student who is involved in the campus life will enjoy the experience and remain in college while achieving academically (Hutley, 2004).

On-campus activities and organizations have been noted by administrators as necessary parts of the campus community (Lambert, 2004). Faculty members' view on campus activities is seen as another way of educating students that is different from the traditional classroom setting (Fry, 2005). In these settings, officials from the student activity center can learn to present educational opportunities as well as fun activities for the students (Lambert, 2004). Learning communities are other avenues where students have opportunities to become involved on campus by connecting with peers, faculty, and staff outside classroom settings (Banta, 2001). At Northern Arizona University (NAU) the belief is that student involvement on campus is equal to retention (Nason-Saltonstall, 2004). The Student Affairs division at NAU has implemented a Datamart which is a set of data that examines whether one set of students who partake in the student affairs programs do better than those who do not take part in the student affairs programs.

The eight services included in the study were: (a) Residence Life, living on campus, which raised retention by 69.6%; (b) LACs, being tutored in one of the Learning Assistance Centers - increased retention by 71.2%; (c) Health Center, utilizing healthcare

services - increased retention by 72.3%; (d) Afterhours, attending late-night social activities - increased retention by 72.7%; (e) Campus Recreation, participating in recreational activities - increased retention by 74.9%; (f) Career Services, attending a career fair or participating in a one-on-one career counseling session - increased retention by 76.5%; or (g) participation in any number of other career planning activities increased retention by an undetermined amount; and (h) Student Clubs, belonging to a student club or organization - increased retention by 86.3%. Freshmen who participated in one of the seven Student Affairs services had a 67.4% higher first-year retention rate than the overall NAU cohort (Nason-Saltonstall, 2004).

According to an ACT Policy Report (2004), there are several factors which contribute to the success of the university student. These factors work together to help with student success, one major factor being student involvement, which seems to positively affect retention and student GPA (Roman, 2007; Thomas, 2005). Thomas (2005) continues his belief in the fact that participation in-group activities is the leading factor in engagement and that teamwork offers collaboration, cooperation, and accountability. Previous research has found positive relations between engagement and achievement, however; it found that involvement does not always translate to engagement (Kayatin, 2005). For instance, a student may participate in a group, which takes away too much of their time, and is not in any way supportive of their school university mission of educating all students served (Upcraft, Gardner, & Barefoot, 2005).

The relationship between student affairs and academic affairs is an important one in that it provides an opportunity for students to be educated from several critical

departments of the university (Fry, 2005). In some ways, it has become clear that university officials have not determined what is considered to be too involved for the students (Bloom, 2006). Some institutions find themselves adjusting to students rather than students conforming to the institution. There are questions that arise out of this dilemma: (a) When does too much involvement become the downfall of the student, and (b) what amount of time is considered to be too much toward campus involvement. Through research, there have been some indications as to the relationship between campus involvement and retention however, it is not clear whether there is an adverse effect on involvement (Pascarella & Terenzini, 1991). This information provides insight into what students learn in a college setting above and beyond class instruction. Student involvement on-campus life is important, and if carefully directed does not take away from any important student learning (Holland & Andre, 1987). Involvement creates an environment that cultivates communication skills and helps in the area of professional development while fostering the ability for group dynamics and positive academic outcomes (Trevas, 1996).

In addition, university professionals cultivate environments that provide the necessary platform for students to learn as well as become involved, hence allowing students to reach desired and attainable goals (Papanikolaou, 2003). University officials need to realize that educating the student is the top priority of the institution and must ensure that balance is achieved in student involvement to maintain or improve retention (Hutley, 2004). Being involved and in some cases overly involved has an effect on a student's GPA and educators must be careful to note this relationship. Geocariss and Goad

(2004) stated, “Student activities offices should be in collaboration with other learning opportunities for students”. More supportive actions will be taken once university professionals in both academic and student affairs realize the impact of student involvement in a total capacity involving physical, mental, emotional, and financial stability (Geocaris & Goad, 2004).

Educators on a collaborative effort need to realize the true diversity of the student body and cater in such a way that needs are met evenly among the different demographics of the campus community. In so doing, realizing that traditional students may be seeking different outcome goals academically and socially compared to nontraditional students (Goad, 2004). Involvement motivates a students’ desire and has the effect of ensuring the student stay in school and ultimately graduate. However, the dilemma is making sure that students do not overindulge themselves in extracurricular activities (Ravitch, 2003). In many cases some students tend to become involved with non-essential campus activities and administrators must ensure that these students come to realize the business of education is the first priority (Johnson & Renwick, 1983).

Armentrout (1978) said it best over 25 years ago: “Activities have great educational values, but we should revamp and reorganize them so that they make a more positive contribution to the objectives of a college education” (Armentrout, 1978 p. 1). In some instances, it is the requirement of university officials to implement quality programs that nourish educational experiences and positive engagement opportunities for the exceptional millennial students (Armentrout, 1978). It is extremely important to identify when students become too involved in campus activities and neglect academic

responsibilities so that measures can be taken to avoid critical failures (Evans, Forney, & Guido-DiBrito, 1998). In an effort to be effective, engagement among students and cost-effectiveness among university departments may need to be streamlined into smaller more focused events (Upcraft, Gardner, & Barefoot, 2005).

Overall academic growth of the student is the responsibility of the university. Events and programs must be so ordered in such a way to ensure that the process happens as smoothly as possible (McGrath, 2002). McGrath suggests, that if students become too involved with campus life and do not do the things necessary to pass exams the university has failed at its duty. A complete student is one who has learned to balance both academic and involvement (Hutley, 2004). To have success at these processes institutions must realize when it's too much and when departments must scale back on larger events and set up a significant program to achieve different academic involvement results (Evans, Forney, & Guido-DiBrito, 1998). This method will improve institutional effectiveness and efficiency when it relates to student involvement. University professionals are always looking for ways to enhance collaboration between academic affairs and student affairs components (Lambert, 2004). All collegiate professionals especially among departments must come together to produce a student that is complete and ready to participate in post-college completion initiatives (House, 2000).

Faculty Instruction in Preparing Students

For learning to occur in a classroom environment, there must be a connection between teacher and student that relates not only content knowledge but relation must occur individually and collectively on a greater social interaction of the demographics of

students being taught. Several studies have indicated a positive impact among African American Students where interaction between student and teacher was viewed as a positive occurrence (Moody, 2004; Littleton 2001). One author suggests when teachers are able to bridge the gap between the students, social language, and the mathematical language there is a communication that takes place and provides a foundation for students to acquire mastery of content knowledge (Vygotsky, 1994). In addition, technology usage by teachers could play a significant role in the learning environment where technology and pedagogy can provide a dynamic approach to learning. A reliable predictor of student learning outcome potential and achievement can be assessed by teachers' level of technology implementation (LoTi). There have been are several frameworks created to measure technology integration in the classroom.

According to Joshi, Pan, Murakami, and Narayanan (2010), who implemented a study on technology integration into the classroom instruction for students to determine teachers' perceptions, 65% of U.S. teachers said it motivated the students. Berkeley-Jones (2012) examined teachers' LoTi self-ratings and student Texas Assessment of Knowledge and Skills (TAKS) scores, finding no difference between teachers' LoTi scores and student mean scores on ELA and math TAKS. Alfaro (2008) established a relationship between teacher LoTi scores and student scores in language arts but found no difference in math and social studies. However, Truett (2006) found teachers' LoTi scores had a significant impact on the math scores of students. Thus, LoTi scores have allowed educators and schools to keep in sight the effectiveness of using technology in the classroom as an instructional tool to meet the needs of 21st century learners.

Chapter Summary

What we have come to realize from the research is that no single theoretical perspective or framework is capable or comprehensive enough to identify all the contributing factors that affect successful academic performance by students. It is also important to note that even though much of the work above relates to student success there are other important factors in this process. The academic background of the student lays a foundation for which students begin to thrive leading to competence in academic ventures. A major consideration in this process is the pre-college experience of those wishing to become a part of the higher education institution and occurrences received within the collegiate environment. Also, vitally important is the educational attainment of parents and financial resources available to help where limitations of grants and other financial contributions end. Once students are able to overcome initial barriers there still exist challenges of integration into the college setting which entails all aspects inclusive of living arrangements, classroom involvement, out-of-class involvement, and interaction with other students, faculty, and other significant university personnel. This adjustment seems to have the most effect on students who are often first-generation or are a part of an at-risk population.

Another important aspect of the transformation is the adjustment to the use of the various type of instruction technologies in the process of student engagement and academic learning. Students who have a strong technical background are far less startled at the requirements associated with the use of technology. Finally, students must believe they are capable of succeeding academically in every aspect and have confidence in their

abilities to master academic content that may be new or difficult. One thing that cannot be neglected is the part the institution plays in the whole scheme of things which is to provide space and resources to actively engage students and provided beneficial learning and developmental experiences (Kuh et al. 2005b; Volkwein et al. 2000). This collaboration needs to be a careful process that utilizes learning experiences in the classroom environment and social involvement outside the classroom while maintaining appropriate balance throughout the experience (Bailey & Alfonso 2005).

CHAPTER III

METHODOLOGY

The method employed to undertake this study used an interview interpretative style approach to gather information from African American Science Technology Engineering and Mathematics (STEM) students who had successfully passed college algebra and/or pre-calculus with a grade of A or B. The investigation sought to gain understanding from participants' experiences as to what constituted success in mathematics and how the identified factors contributed to that success. In that regard, mostly non-numeric data was created from these study participants about strategies and procedures implemented that create an arena where mathematical success could be achieved. The overarching research question for this study was: What strategies do African American STEM students rely on to succeed in university-level mathematics such as College Algebra and Pre-Calculus, and by what means is this accomplished?

Research Design

To address this research study, a qualitative research design was employed using a phenomenological research analysis. According to Creswell and Poth (2018), posits that phenomenological research analysis allows for valuable data to be gathered from research participants regarding their experiences relating to a concept. In addition, when analyzing qualitative data the primary goal is to allow the accurate experiences of the participants to come to the forefront. Husserl (1970) further states, “Pure phenomenological research is

concerned with description rather than explanation and is based on a perspective that is without preconceived notions or hypotheses.”

Setting, Population and Sample

Blue University (pseudonym) was the setting for this study. Blue University is located in the southeastern region of the United States with a student population of a little under 8000 students in 2018 consisting of both undergraduate and graduate students. The undergraduate student-faculty ratio is 17 to 1. The population for this study was STEM students who had completed college algebra and/or pre-calculus with a grade of A or B grade.

Even though a C grade is considered passing for all universities this study choose not to consider the C grade based on a study conducted by Georgia State University which suggest that when students obtain a C or D/F grade within their first major-related course the likely hood of staying in that major or being successful in other major courses was considerably low 39.5% and 8.3% respectively. On the other hand, when the grade was an A or B grade the likelihood of staying in that major and being successful was 70% and 67% respectively. Hence, when an A or B grade is achieved in the first major-related course students are far more likely to be successful in their intended major. This result was seen across all majors within the university.

From this population, a sample of eight to 15 participants was randomly selected through an excel spreadsheet with the purpose of seeking saturation within that number. Saturation in qualitative research is where no new themes or codes are observed in the data (Guest, Bunce, & Johnson, 2006).

Instrumentation

The data was obtained through face-to-face or telephone interviews using a standardized series of open-ended questions derived from the survey questionnaire (see Appendix B). The questionnaire was used as a guide and contained 14 questions. Questions were selected in any order and answer responses were further questioned for additional clarity. Notes were taken and voice recordings were taken for later review and analysis. This initial questionnaire, with written permission and modification, was used from a dissertation study titled: *African American Male Students' Perceptions of Factors that Contribute to Their Academic Success*. This study was conducted by Dr. Gertrude Rolland. Dr. Rolland was contacted (see Appendix C) and gave permission via e-mail (see Appendix D) to adjust the instrument as need so long as credit was given. Additionally, before use, testing for validity, and reliability was performed to limit flaws in the questions. In addition, a demographic questionnaire was given to help in the identification and selection of acceptable study participants (see Appendix E).

Procedures for Data Collection

According to Schwandt (2000) and Patton (2002), the interview approach allows the researcher to understand two major concepts; how the “inter-subjective world is constituted of the participant and what is in the minds of those who have experienced the concept. Also, with the interview method, there is a greater degree of flexibility which allows best for the multiple usages of data collection approaches to answer the questions of a research study (Cavaye, 1996). Data were collected through two methods, (a) a demographics questionnaire and, (b) a specifically design open-ended survey

questionnaire. Questions on the demographic survey included age, gender, parental income status, and first-generation identity. In addition, students responded to a series of open-ended questions from the survey questionnaire. Qualitatively, question responses from the open-ended survey questionnaire were used to glean information to provide context and structure from the student responses and seek answers to the study questions.

With the proposal and IRB approval, a permission letter was sent to the mathematics department head to provide information about the study and seek permission to communicate the study with individual classes. Once approval was granted, the researcher met with selected faculty members to request permission to speak with their respective classes. During the initial meeting with the classes interest in the study was gauged through an initial demographic questionnaire. Once the demographic questionnaire was collected it was reviewed and number stamped to keep any significant information private. After which a random number system was set up in order to select between eight to 15 participants for the study. Selected participants were notified, and appropriate times were selected for the interview process to occur.

On the initial meeting with the selected participants, the time commitments and privacy issues were addressed before the interview was started. The interview process took between 30 to 45 minutes to complete. In addition, notes and recordings were taken to provide further clarity to responses obtained from study participants. After the initial interview, a second member checking interview was conducted to ensure participants agreed with the interpretation of the information gathered.

Data Analysis

After the interviews are completed the data was transcribed with notes added to ensure completeness of thoughts; organized with study narratives and cross analysis then finally coded to find patterns and themes that sought to address the study questions and find context within the existing literature. According to Creswell (2007) and Creswell (2012a) coding is used to take small pieces of data and provide meaning from text and create themes in data analysis. The data collected was analyzed using Dedoose, which is a qualitative software that helps with the process of qualitative research analysis. To ensure complete privacy pseudonyms were used for all the participants. The Creswell Data Analysis spiral approach was used which sought to provide understanding at various levels of the data analyzing process. All information collected was kept secure through password protection and lockbox facility to provide confidentiality to the study participants.

Chapter Summary

The qualitative approach design, using a phenomenological analysis was used to allow the researcher to collect data through an open-ended interview style questionnaire to evaluate the lived experience of African American students who have had success in freshmen level mathematics courses such as College Algebra and Pre-Calculus. The questionnaire consisted of 14 questions which were flexibility and were used to create a starting dialogue into the perceptions of the selected study participants. The Creswell Data Analysis Spiral approach was used because it provided a well-balanced systematic process to help with understanding the qualitative data analysis process and created a

context for providing themes and meaning to dialogue. The data was transcribed, organized, and coded to seek out patterns and common themes among study participants. Finally, to adhere to the necessary security of data, all collected information from the study participants was stored inclusive of transcriptions, audiotapes, and private data in password-protected formats and secured locked boxes.

CHAPTER IV

RESULTS

This chapter details the results and analysis of the data collected during this study.

Ten STEM students were interviewed to understand the perceptions of math success by these students, the pre and in-college experiences that have contributed to their success in math, their belief in self and motivational factors, the importance and uniqueness of the HBCU experience that has impacted their math success, and the stigma that society portrays towards African American women in STEM-related disciplines. These students were currently taking a math class and all of the students had GPA scores over 2.8. See Table 1: Participant information.

Table 1: Participant information

Participant	Major	GPA	Classification	Current Math Class	First Gen.	Income level High/Low
Shelton (M)	Mech. Engineering	3.50	Freshman	Pre-Cal II	Yes	Low
Toran (F)	Chemistry	3.27	Sophomore	Calculus I	No	Low
Christian S. (M)	Civil Engineering	3.06	Sophomore	Calculus I	Yes	Low
Tamera (F)	Biology	2.80	Senior	Calculus I	No	High
Jalyn (M)	Health Sciences	3.41	Junior	Pre-Cal II	No	Low
Destiny (F)	Biology	3.41	Freshman	Pre-Cal II	No	Low
Sean (M)	Mech. Engineering	3.13	Freshman	Calculus I	No	Low
Harmonie (F)	Health Sciences	3.50	Junior	Calculus I	Yes	Low
Christian B. (M)	Biology	3.24	Freshman	Pre-Cal II	Yes	Low
William (M)	Chemistry	3.79	Freshman	Pre-Cal II	No	High

Through the interview process and by the data collected the research questions were addressed:

- 1) How do African American STEM students perceive success in mathematics at a Historically Black College and University (HBCU)? (RQ 1)
- 2) How have experiences of African American STEM students (e.g., past and present) contributed to their success in mathematics? (RQ 2)
- 3) To what extent being successful in mathematics contributed to overall self-efficacy and motivation of African American mathematics students? (RQ 3)
- 4) How has attendance at an HBCU provided opportunities to meet the needs of African American students in STEM related discipline? (RQ 4)
- 5) Is there a perception of African American STEM female students that gender plays a role in their involvement in STEM related disciplines and success in undergraduate entry-level mathematics courses? (RQ 5)

Table 2 displays the connection between the research questions and the questions in the research protocol.

Table 2: Interview Protocol

Research Question	Interview Questions
RQ 1	2) What does it mean for you to be successful in mathematics courses? 3) What strategies have you implemented that have caused you to be successful in mathematics courses?
RQ 2	1) Tell me about yourself (background, upbringing, family structure, schooling, etc.) 4) What challenges and/or obstacles were created while taking mathematics courses? 5) Discuss significant role models (parents, teacher, family, friends, etc.) that may have impacted you being successful in mathematics. 7) Talk about how technology was used in your secondary schooling math courses and the role it plays in your post-secondary mathematics courses.
RQ 3	6) What motivated you to select one of the STEM disciplines knowing that mathematics would play a significant role? 8) Does being involved with university clubs, sports, and/or organizations impact your success in mathematics, if so in what ways?
RQ 4	9) Being African-American what and how have attendance at an HBCU contributed to your math success and overall student satisfaction.
RQ 5	10) What are your thoughts about female African American students choosing STEM disciplines and having success in mathematics as their male counterparts?

To undertake this study, the researcher went to five math classes to present this study to the students and identify interested students. An initial demographic analysis was provided to students wanting to participate to determine if they would be eligible.

A total of 51 students filled out the demographic profile however of that number only 35 met the criteria for the study. From these 35 students, a random sample drawing in Excel was used to select 10 students who would be interviewed for the study.

Interviews were scheduled and all interview sessions were held in an office in the mathematics department. Interviews were recorded and notes taken to capture the entire interview experience watching for any emotions or "body language".

To grasp a better understanding of all the data collected the researcher listened to the audio recording. After which the interview transcriptions were read and reread to get a sense of cohesion and generalization about participant interviews. Once the summaries were completed, the researcher shared the information with the participants to check for accuracy and if necessary revision. The final participant summaries are described below. After the summaries, the analysis of categories and themes found will be presented with participants' reflections. Initially, there were 129 codes found through the in-vivo coding process, which used the participants' own words as initial coding after which these codes were grouped by similarities and then reduced to 29 codes which were then group together and places under categories that produced themes to address the five research questions. The summary of the findings is displayed in Table 3, after which the participants' summary is relayed and the finding relating to the themes and categories.

Table 3: Themes finding summary

Themes	Categories	Description
Perceptions of Math Success	▪ Maintaining academic excellence ▪ Linking to future career success ▪ Overcoming	African American STEM students knowing and understanding what is required to achieve success in mathematics but also Understanding the

	- success obstacles - Success factors - Success strategies - Support systems - Technology Implementation	connectedness of other related factors to achieving that math success.
Pre-College Experiences and Math Success	- Early academic beginnings - Expectation of high school vs. university - Family expectation - Family support - Extracurricular curricular activities	The importance of having the necessary tools and foundational math experiences along with the social and community support of family and other educational partnership
Self-belief and Motivation connected to Math Success	- Believing in one's ability - Future math success - Having balance with extracurricular activities - Math success motivations - Promoting a lasting legacy for African American women STEM success	African American STEM students' ability in self to not just comprehend math content but master the content. This allows for students to be motivated to achieve the desired outcome.
The HBCU Experience and Math Success	- HBCU effect on math success - Help from professors from HBCU institutions - Importance of the HBCU connections to math success	The impact of the HBCU experiences and influences that have provided a nurturing environment for African American students
Women in STEM and Math Success	- Access to STEM for African American women - African American women in STEM as role models - Society's view of where women	How African American women in the STEM fields are viewed by themselves and others.

should be

Participant Summary One (Shelton)

Shelton is a freshman mechanical engineering major born and raised in Nashville, Tennessee. He is a first-generation student and holds a 3.5 GPA. During this interview, Shelton was enrolled in Pre-Calculus II and had received a B in his previous math class Pre-Calculus I. Shelton plans to become a mechanical engineer and working for a major engineering firm. Shelton attended McGavock High school and got all As and Bs in the four math classes he took. In high school, Shelton played several sports including basketball and track and field. Even at times during sports training, Shelton would look for ways to insert math in the training sessions. One time sitting on the side he calculated his running speed by observing the distance he ran and the time it took to reach that point. Shelton has a twin sister who he considers an inspiring role model and credits some of his academic ability to the influence she had on him as a young child growing up. His parents were hard on both of them and always challenge him and his sister to be the best at whatever they decided to accomplish. Expectations by their parents were only to get As and Bs and consistently to be on the Principal's list for students who were excelling academically. Shelton noted that he and his sister both had perfect attendance. Understanding the difficulties he had with math always pushed him to strive for excellence by seeking help prior to needing the help. He studies daily for several hours and attends tutoring as much as possible during his busy schedule.

Not being able to use calculators in his math courses, Shelton relies heavily on mental math and note-taking to be successful in his math courses. He also attends office

hours with his professor when he has difficulties with some of the online homework assignments. On several occasions, he and some of his friends would go to the library and have study sessions, especially before major exams. Attending an HBCU institution was important for Shelton because it allowed him to see others like himself excelling in their various disciplines. He also noted the importance of having a professor that looked like him and in some ways understood the challenges he faced as an African American male student. When considering his female counterpart in the STEM discipline Shelton believes that they can do whatever they desire. In no way does Shelton think that African American females are not as smart as African American males and notes this by saying they are the ones who usually ask the most question in class and provides most of the answers when asked questions by the professors. He believes that the male students do not want others to think they are not smart. One thing he would change about himself is the way he often procrastinates because he knows that when you procrastinate you usually are not able to accomplish good things.

Participant Summary Two (Toran)

Toran is a sophomore Chemistry major born and raised in Birmingham Alabama, "Roll Tide Roll". She is a second-generation college student and holds a 3.27 GPA. At the time of this interview, Toran was enrolled in Calculus I and had passed her previous math course Pre-Calculus with an A grade. Toran plans on becoming a chemical engineer, she recently thought about becoming a medical doctor. During her early school years, Toran was placed in many advanced placement courses and did very well in all of those classes placing on a path that would lead to her current major. Toran came from a

single-parent home in which she grew up with her mother. At a later stage in Toran's life, her father would play a significant role. She has five siblings two from her mother's side and two from her father's side. She notes they all get along pretty well. Toran's mother was able to continue her education beyond high school and acquire an Associate's degree. Her father works in Information Technology (IT) and has acquired numerous certificates in the field.

At present Toran is not on a scholarship but is interested in applying for one at a later date. When thinking about what constitutes having success in mathematics it is mainly understanding concepts and being able to apply the concepts to problems. During many of the math classes, she has taken the greatest struggle that she has faced is having to remember a lot of information and formulas. Even though this was a major struggle for Toran she always relied on people who were able to help her in these areas. Toran noted that two of the persons who impacted her life in the area of mathematics was her mother and several of her math teachers. Besides math, Toran says that Chemistry is one of her favorite subjects. Her motivation for becoming a STEM major is due to the fact that she loves to see people with beautiful smiles and healthy teeth. Toran is really careful to pay attention to her academics but she always makes time for other extra-curricular activities. Toran was on the track and field team and the dance team in school. Presently she is a part of the American Chemical Society at the university. She made sure that these extra-curricular activities never interfered with other important tasks related to school academics. During high school there was never a problem with the usage of the calculator but while attending college that was taken away and because of it, she became better at

mental math. As far as technology is concerned it has its place and she believes that having the textbook online and the homework online is not a major issue but she does have concerns with some of the presentation of the material. In Toran's major, she finds some difficulty with the lectures because many of the lectures are not native English speakers and this makes taking notes for later study sometimes difficult. One major thing that she loves about being at an HBCU institution is the assistance that the professors are willing to give you as a student. She does not feel like she is just a number. Her view of African American women in the STEM field suggests that women can do anything they put their minds to do. She believes society sometimes limits women to only caregivers. She also believes the university professors should be more open to different styles of teaching methods.

Participant Summary Three (Christian S)

Christian S. is a sophomore civil engineering major who was born and raised in Lansing Michigan. Christian S is a first-generation college student and has a GPA of 3.06. At the time of the first interview session, Christian S. was enrolled in Calculus 1. In his previous math class college algebra, he earned a B grade. Christian S., plans on becoming a civil engineer in a large engineering firm. Christian S. has 17 siblings, 8 brothers, and 8 sisters; he is the baby of the siblings. During his early years, there was a lot of pressure to keep up to standards with whatever great things they were doing. Despite having so many siblings Christian S. decided to pursue his path. He is the only sibling to leave the state of Michigan and pursue an engineering major while attending an HBCU institution. Neither of his parents attended an HBCU.

Christian S. is not on a scholarship but has a scholar rate which helps with his tuition if he keeps his GPA above a 3.0. When questioning Christian S. about what he believes to be successful in math he notes that having the ability to overcome challenging problems and helping friends who are struggling with the same situation. Overcoming challenges in math was not easy but his parents always provided support to assist him in areas where he encountered struggles. On several occasions, he was supported by a tutor who timed him when he was working on his math problems to increase his critical thinking speed. Now in the university setting, Christian S. continues to work on problems repeatedly until he is able to master the concept. During testing Christian S. usually has test anxiety so he uses breathing techniques to settle himself before taking exams. He also believes the teachers at Blue University provide the needed support to students who are experiencing difficulties. He notes that even though he has failed several tests along the way he always manages to pass his class. In difficult times Christian S. looks to two close role models, his brother and his mother. Christian S. choose engineering as a major because he always loved working with his hands and during his senior year in high school his school had a visit from the transportation department and that made clear what he wanted to do, become a civil engineer. The following summer he worked with the Michigan Department of Transportation and got introduced to the many aspects of civil engineering he had no idea about. During his math classes in high school, Christian S. was allowed to use calculators often but this often was a shortfall because the teacher never really worked out the problem for the students.

In college, it is much different because you are not allowed to use computers at some institutions so they usually show every step in the process of trying to solve problems. In high school, Christian S. ran track but he also played football because all of his brothers played football. He believes he was best at track because he enjoyed the math involved with for example "tempo runs". Attending an HBCU institution was a goal of Christian S. because he wanted to see other students who look like him excelling academically. He believed many of the students had similar backgrounds and faced some of the same circumstances growing up. He believes African American women and men are both able to successfully meet any challenges they face but sometimes society tends to try and limit how far women can go and what they become in their profession.

Participant Summary Four (Tamera)

Tamera is a senior Cell and Molecular Biology major who plans on becoming a dentist. Tamera was born and raised in Saint Louis, Missouri. She is a second-generation college student and the only student who did not have a GPA above 3.0; her GPA was 2.8. This semester Tamera was enrolled in Calculus I and had recently passed her last math class Pre-Calculus I with a B grade. Tamera went to Clayton School District and attended one of the top schools in the district. She had two sisters and both parents who previously attended college. Tamera mom holds a doctorate degree in English; unfortunately, her father never finished college. Her mother works in education and has held several high profile positions including, vice-principal, principal, and superintendent position. Her dad works in the post office as a supervisor and together they have a family business. For Tamera success in math is twofold, both getting good grades and

understanding the material. Tamera often relies on tutoring for assistance whenever she finds herself struggling in any of her classes especially math. She dislikes the way math is taught using the technology and prefers the “old school way” which is using pencil and paper to submit homework assignments.

Significant role models who have played a vital role in her math journey are her mother along with several of the math professors both in high school and in college. She wanted to become a dentist because she admired her dentist as she was growing up. Her dentist was an African American female who took Tamera under her wings. Attending a technology-based high school environment where everything in math was done with a calculator was easy, at Blue University where calculators are not allowed has been quite a transition. Tamera participated in several clubs and organizations while in high school. Attending an HBCU institution has made Tamera comfortable as she does not receive any belittling from any of the professors. Tamera believes that African American women can do STEM and she has seen many in her schooling who have paved the way for her. Her desire for wanting to become a dentist is to help people who have limited or no resources for dental insurance. Tamera is resilient and has confidence in her abilities and knows one day she will become a dentist.

Participant Summary Five (Jalyn)

Jalyn is a junior Health Science major who was born and raised in Montgomery, Alabama. He is a second-generation student who holds a 3.41 GPA. Jalyn is taking Pre-Calculus II this semester after having passed his previous math class Pre-Calculus with an A grade. Jalyn's future career goal is to become a medical assistant in a critical care

hospital. Jalyn has two siblings a brother and a sister. His brother is a freshman at Tuskegee and his sister is a ninth-grader. Both of his parents are college graduates earning bachelor's degrees. Jalyn is the oldest sibling and attended Brewbaker Technology Magnet school and was a part of the medical academy. For Jalyn being successful in math is about understanding the material and being able to apply the learning to other subject matters. Two of Jalyn's biggest strategies for being successful in math are repeatedly doing problems and seeking assistance often. Technology is great in many ways but Jalyn thinks sometimes it makes you learn at a slower pace and believes it should be used with different platforms. During many of his math classes, professors seem to relate the same information in many different ways and for Jalyn at times this became very confusing and caused him to struggle unnecessarily.

Jalyn credits his grandmother for the inspiration of going to college and becoming a Health Science major as she was the first person in his family to attend college. His grandmother was a math major and the oldest of ten siblings. In addition during high school, Jalyn was a part of the medical academy and had opportunities to visit hospitals and rehabilitation clinics where his love and passion heightened. Jalyn receives the scholarly rate and is also on an athletic scholarship. He appreciates the way professors integrate technology in the classroom environment and noted in high school it was not used much even though they were often allowed to use calculators. Playing sports was an area Jalyn excelled, and in fact, he played many sports while in high school including cross-country and basketball. Here at Blue University he continues to excel in sports and is a long jumper on the track team. The university provides tutors for athletes so usually

his playing sports do not affect his academic performance or his math and science classes. His desire to attend an HBCU is to gain recognition from those of similar backgrounds and to have the interaction of faculty who has a similar background as he does. He believes that the classes are smaller and so it provides more interaction with the professors to get to know them and they get to know him. He believes society sees the place of the women is to be subservient but thinks that many of his peers who are African American females are performing well in many of their classes and believe the very opposite.

Participant Summary Six (Destiny)

Destiny is a freshman Biology major born and raised in Nashville, TN. Destiny is a second-generation college student who holds a 3.41 GPA. In her previous math class, College algebra, Destiny earned a B grade. She is now enrolled in Pre-Calculus II and plans on working towards an A grade. Destiny plans on becoming a dentist and working for an organization that supports disadvantaged communities. Destiny knew all her life that she would have to make something of herself because her entire life she grew up around educated people. Both grandparents have their doctorate degrees. Her mother and several of her aunts all have master's degrees. Her mother always stressed to her to go after really good grades and understanding the material given in each class. She is an only child who grew up with her mother. Her father passed when she was just three years old. Being successful in math for Destiny is having the ability to apply what you are learning to real-life situations. The biggest thing that Destiny would tell all of her classmates about becoming better in math is time management and repeated problem working. For

Destiny, tutoring is not something optional, and so she finds her way to tutoring almost every day and enjoys study time with her peers.

Sometimes, Destiny is challenged by the pace of material covered in the math class and believes the professors should not move so quickly over the material. However, to combat this she takes good notes and revises the material every day, and brings back questions when something confusing arises. Destiny received a scholarship through a third party source and must maintain at least a 3.0 GPA. The scholarship award is around \$4,000 per year. When needed additional help with math, Destiny seeks the expertise of her granduncle who is a mathematician at Alabama State University. She considers him to be a great role model. Her major desire to pursue a STEM discipline, particularly biology was to become a dentist. Technology in high school was not a driving factor but at the university setting you have to be tech-savvy or you may get lost in the shuffle as noted by Destiny.

Destiny, in her first math class, realized the seriousness of what was ahead and made the decision she had to become very serious about college, or she would fail and probably let her family down. Destiny is an avid basketball player and played the sport from grade seven and believes it gave her structure for school and life because of the discipline required to become good at basketball. She believes if sports, clubs, and organizations are not done in their place they can become a hindrance to academic progress and excellence. Destiny never thought about attending any other institution besides an HBCU. She believes the HBCU experience provides you with a historical perspective, class sizes are smaller, interaction with peers and faculty is more natural and

there is little judgment or criticism among each other. Everybody wants to help.

Destiny's mother is also an HBCU alumnus. Destiny sees females as nurtures but still believes African American women can excel in the STEM arena even though they are very limited in the profession.

Participant Summary Seven (Sean)

Sean is an 18-year-old freshman mechanical engineering major from Saint Paul, Minnesota. Born and raised in Nashville, TN. Sean is a second-generation college student who holds a 3.13 GPA. In his previous math class College algebra, Sean earned a B grade. He is now enrolled in Pre-Calculus I and plans on working towards an A grade. Sean desires to become an engine and vibration specialist in the mechanical engineering environment. Sean's early schooling placed him in predominately STEM-oriented schools. Many of the classes he took were centered on some math and engineering concepts. Sean's is not the first to go to college but he has a long lineage to look at, his mom and two sisters attended college but his father and oldest sister did not. Sean received a scholarship and has to maintain a 3.3 GPA to keep the scholarship. For Sean success in math means having the ability to take the information you are learning and applying it to some math problems. In addition, being able to apply what you learn in math to other STEM-related classes. Sean considers himself to be a very proud person so seeking help when needed was very difficult at first. However, after speaking with peers who were receiving help through tutoring Sean decided to give tutoring an opportunity. Since then he has always used the tutoring services. Sean and his sisters always competed in math to see who would receive the highest grade and because of this, they all became

better in math. He credits his mom for instilling this drive for excellence in each of them. Sean did not have to look far for role models because he had many and one important figure was his cousin who is a civil engineering major. He and Sean always played together when they were younger and use to break toy cars apart and then rebuild them. This made Sean love the mechanical aspect of cars and during these times decided to become a mechanical engineer. Sean's high school was a school that relied a lot on technology. All the students were given iPads for daily academic use.

However, in the university setting, there is limited access to technology especially without having the ability to use graphing calculators for tests or exams. Sean has no problem with having the textbook and homework online but understands it may be a problem for some of his peers. Even though language barriers between professors and students may exist if you pay attention you are able to understand the material presented. In the event that fails you can always seek help from the resources at the university. Sean participated in both basketball and volleyball in high school but does not plan to play any sports in college. Sean believed his playing sports in school distracted from his academic pursuits and placed him at a disadvantage when starting his college journey in mathematics. The support provided within the HBCU environment is second to none especially for a student like Sean who needed to see others like himself excelling. He always receives a push from the professors urging him to always set the bar high. Sean's teachers were predominately white in middle and high school coming from Minnesota so having the opportunity to see teachers like him is something that daily inspires him to excel. He believes that both African American males and females can excel at anything

but believes challenges exist that put a greater burden on the females African American STEM students from wanting to pursue majors and careers in that field because of chauvinistic attitudes by men. Sean would like to see the college professors teach in ways that resemble the best understanding of the students in the classes and believes that even though math can be a very difficult subject at times but with the right teaching methods and a strong desire to learn you can become very successful in any math class.

Participant Summary Eight (Harmonie)

Harmonie is a junior Health Science major who was born and raised in Nashville, TN. She is a first-generation student who holds a 3.50 GPA. Harmonie is taking Calculus I this semester after having passed her previous two math class College Algebra and Pre-Calculus earning B grade in both classes. Harmonie desires to become a Physical Therapist in a local hospital. She attended a private middle school and then later completed her high schooling by attending a public school system in her hometown. Harmonie comes from a very close-knit family having an older brother and a younger sister. Neither of Harmonie's parents attended an HBCU but did attend community colleges. Harmonie is not on a scholarship but intends to pursue some at a later time. For Harmonie being successful in math is more than getting a certain grade it's about understanding the material and being able to apply it and solve many kinds of problems. Beginning in elementary school, Harmonie implemented many ways of learning new math information including singing songs that helped with memorizing the timetables and formulas for math equations. During her current math classes, Harmonie at times struggles because she is used to be allowed to use a calculator but at the university where

she takes her math classes' calculators are not allowed for exams or tests. In addition, she often struggles with the language of the instructors who are not native English speakers.

Harmonie thinks that technology is beneficial in some instances but does not like that the textbook and home are all online. For Harmonie, this seems to distract from her mental abilities as she is used to working out math problems with pencils and notebook paper. Harmonie has always relied on the push of her family members who always played a major role in Harmonie's life even though growing up she was very shy. While on the campus her mother also insisted that she seek tutoring support whenever needed. Harmonie believes that if you provide support to others in their time of need they will reciprocate the favor and this led her on deciding to become a healthcare worker.

Throughout her early schooling, there was limited access to resources like a textbook but she was able to rely on the calculators while doing the assignments on the worksheets provided. She believes the teachers could have provided more support for them, but because of the limited resources felt like their hands were tied. Harmonie while in high school participated in sports for three years running track and competing in long jump competitions. Her favorite distances were the 100 meter and 200-meter races. In addition to sports and academics, Harmonie also found time to partake in the Rotary Club and at the university level become a member of the Honors College. For most of Harmonie's life, she attended predominately white schools, and attending an HBCU institution is something she decided while in the ninth grade. She wanted to experience the talent and greatness of others who were from similarly disadvantaged backgrounds and interact with professors who understood some of her stories. Society's view of

women and men does not portray them often as equal hence the disparity in the pay however she believes that both African American women and men can perform well in mathematics and become successful STEM majors. Harmonie initially struggled in math but because of tutoring she has become really good and understands the importance that math plays in everyday life.

Participant Summary Nine (Christian B)

Christian B. is a freshman Biology major born and raised in Atlanta Georgia but currently lives in Douglasville Georgia. Christian B is a first-generation college student who holds a 3.24 GPA. In his previous math class College algebra, Christian B. earned a B grade. He is now enrolled in Pre-Calculus II and plans on working towards an A grade. Christian B. plans on becoming a Medical Doctor, a cardiologist in a rural area. He is from a single-family home and lives with his mother and brother. Before attending college Christian B. attended Chapel Hill high school. Many of his friends and family members have always struggled in math but Christian B. he sees being successful in math as breaking "generational curses" as well as being able to apply what is learned to other harder math classes. When he excels in any of his math classes he feels that he is able to accomplish anything and feels like he is, "just above the world." In his high school, many of the students relied on the use of calculators but not having access to the use of calculators on tests and exams at the university has made some of the classes extremely challenging. To overcome this challenge Christian B. attends tutoring and often finds him and his friends in study sessions. When questioned about the technology usage in his class he prefers having homework assignments and the textbook not online. Inspiration

for Christian B. has to be his fifth-grade math teacher who pushed him to delve into math courses and also one of his colleges' math professors who he says is the "best college-level math teacher ever." Christian B. loves the medical field and understands the importance of having a strong background in mathematics and science. Technology usage in high school and the university seems to be the same according to Christian B. however the difference is not having access to the manual textbook or the usage of calculators during exams and testing. Christian B. played many sports in high school inclusive of football and track; he was also a part of the marching band. For Christian B. the sports he played gave him a sense of organization and helped him with many of his academic classes. He took part in many clubs and organizations including DECA which was a marketing club. Continuing partaking in clubs and organizations he is a part of the tiger tour guides at the university he is attending. Attending an HBCU was important for Christian B. because he wanted to experience what it was like to attend a "black institution." He also wanted to experience the smaller class sizes, peer interaction from persons with similar backgrounds, and professors who were able to understand the demographic perspective.

Participant Summary Ten (William)

William is a freshman Chemistry major born and raised in Nashville, TN. He is a second-generation college student and holds a 3.79 GPA. At the time of this interview, William was enrolled in Pre-Calculus II and had passed his previous math course College Algebra with a B grade. William plans on becoming a chemical engineer or a Chemist in the medical profession, he had not decided at this point. William attended predominately

white middle and high schools William decided he needed to attend an HBCU institution to understand the heritage and academic competence of people like him. He initially went to a catholic school at an early age but by the eighth grade, he was attending Ellis Middle school and the finish his high school career at Station Camp high school. His family support is high and has placed a lot of pressure on him to succeed. His father is a lawyer and so for William, the bar is already set high. He often feels the "weight of the world on his shoulders". He is the older of his two brothers and knows he has to set an example for them to follow. William played many sports in high school but basketball was his favorite, he does not plan to participate in any university sports. He is a part of the honors college and likes the guidance he receives from the college when he needs direction. His father was his inspiration for wanting to join the honors college because he believed it would provide William with uniqueness to build his resume. William believes to be successful in math the professor must relay the information in a way so that students are able to understand. He also believes you should seek out tutoring help when faced with difficult professors who may present a non-native English speaking language to their lecture style. He is a big advocate of tutoring services and believes if he had not used the tutoring services provided he would have failed his first pre-calculus test. Good grades alone are not what is important for William; he would rather understand the material knowing that if you do understand the material good grades should follow. The biggest math strategy for William is constantly reviewing the notes from previous classes and don't think you know everything after leaving the class; do a lot of out of classwork. When asked about role models he quickly says his parents and some of his high school

teachers. After having scored so high on the math section of the ACT and taking many career assessment tests he knew he wanted to pursue a career in the STEM field. During early school technology played a pivotal role in their learning environment and most students had access to Ipads and graphing calculators for use in math courses but at the institution, it is not allowed when taking exams or test, and for William that created a huge problem and major obstacle when enrolled in his math courses. He does not like having the textbook and homework online and sees it as a huge distraction. At the time of this interview, he had not at any point looked at his textbook online. He was adjusting to the homework but noted you have to be very precise when giving answers to the computer. Having success in math has built William's confidence and has motivated him to excel in other classes. He takes the critical thinking approach to all of his other courses. Attending an HBCU institution was a top priority for William who liked having smaller class sizes, professors you can talk too and peers who were coming from similar backgrounds as you. When questioned whose better at STEM African American men or women neither he replied they are both just as good. He thinks society tries to dictate where a woman should be and does not provide the same opportunities for women as men in the STEM field. Despite society's views and notions he believes they can alleviate any gaps and have the same opportunities to succeed as their counterparts.

In the remaining sections of this chapter, the major relevant categories under the five themes that emerged from this study will be addressed. The five themes that emerged from this study include: (a) Perceptions of Math Success, (b) Pre-College Experience and Math Success, (c) Self-belief and Motivation connected to Math Success, (d) The HBCU

Experience and Math Success, and (e) Women in STEM and Math Success. In addition, participants' experiences will be aligned to collaborate with the relevant categories under the five major themes. The five main themes along with the relevant categories will now be relayed.

Perceptions of Math Success

When examining researches it is clear that there exist numerous definitions of what is considered to be mathematically successful. For example, D. Martin (2000) suggests students who are successful in mathematics as, “students receiving high grades in mathematics and expressing positive attitudes about mathematics learning” (Martin, 2000, p. 124). Other authors such as Moody (2004) and Powell-Mike (2001) suggest that those who are mathematically successful have completed calculus sequences courses or other high-level mathematics courses.

Maintaining academic excellence

Several of the participants had scholarships and because of this they were required to maintain above a 3.0 GPA, in addition, there was pressure placed on them by their family to ensure they always strived for excellence. Sean gives the impression of having the weight of the world on his shoulders:

Um for my family, my mom went to college, and two of my sisters went to college. My dad not, and my oldest sister tried, but did not finish. Um, but, coming up I always kind of liked had the weight of the world on my shoulders. I always liked had to stay in school and get really good grades. Mathematics was always my favorite course to do and was always like a family thing to talk about sitting around the table. So that always helped out. I am on a general scholarship and must maintain at least a 3.3 GPA to keep my scholarship.

In similar fashion Destiny also realized the importance of maintaining a high GPA or she would lose her scholarship and have to pay for her schooling as her parents were not able to afford it. She puts it this way:

Um, actually I am on a scholarship and I have to do everything possible to keep it. It's not through the university it's through a third party. They give me, I think about \$4,000 for four years. And I have to keep my grades up. I have to keep like a 3.0. or higher to keep it.

Linking to future career success

Many of the participants knew what profession they wanted to pursue and it seemed like many had this idea from an early age. Even though they understood the difficulties of the STEM field and the many difficult courses their desire to achieve their career goals made them want to excel more than any obstacle. Sean puts it this way:

I would just say, um, STEM, it's a very challenging major to get into, and I know math would play a pivotal role but I'm going to do it because I have goals that I will achieve. Um, I just feel that people need to be focused and determined if they want to get into a major like this, they just got to be ready for challenges, and different things, and aspects associated with this major looking forward to graduating and working in their chosen profession.

Tamera knew the difficulty of becoming a dentist and the odds of not making it but she knows that hard work and determination will allow you to be great at whatever you put effort towards; she talks about it in this manner:

Um, I think that now, seeing females, like being the president of my organization, a lot of people drop out because it gets difficult. Uh, it's a challenge that you have to mentally prepare yourself for because it's not easy. I think, in my years of college, this is probably my hardest semester. Um, and I know it's preparing me for dental school because I find myself saying so many times now that I didn't know school could get this hard. School comes easy to me but now it's just like oh, my goodness. Like, you

have to study. You really have to work more than five hours at this every day, and also studying for my dental exam, it's something that if you wanna do, you gonna do it. So I feel like it's a mind thing. Like, I mean, you talk to dentists now, you can talk to dental students in the area and they'll all say like, "I have breaking points all the time but I stuck with it and became successful at my career."

Overcoming success obstacles

All participants faced difficulties while trying to become good at math. Some of the difficulties were Pre College like losing a parent and while in college like have major test anxiety. Christian S explains it this way:

Um, I faced a lot of challenges as far as tests. I have test anxiety a lot. So whenever you say test, I, immediately go blank. Like, I could know the material, but when it's time to take a test I get very scared. It's just like, I don't want to mess up on this test so I have to be focused. I don't want to overthink it either and get a very bad score on the test and cause my confidence to hit rock bottom. I love the teachers here, but if you don't find the right teacher for you in mathematics, it's not going to be good for you because you may have difficulties understanding the material.

A lot of math students believe they understand the material before they fully allow themselves to be vested in the curriculum these participants were no different and William puts it this way:

The biggest challenge I'd say is just thinking that I know something before I really study it and just being like, "Yeah, I know this, and I'm not gonna study this." I'd say that's the biggest challenge because sometimes I feel like I understand something more than I fully do but then when a test comes up I don't do as well as I thought I should have.

Success factors

Many of the participants wanted to be successful in math and had a desire to do whatever was necessary to achieve that success. The main goal of many of the

Participants was to get really good grades, understand the material in different ways, be able to apply the learning to other relevant courses, and use it to benefit their future in the work environment. Sean presented it like this:

I think to be successful success in math you need to have a basic and in some cases a complex understanding of every aspect of the course content. You must also be able to apply what you learn to other relevant major courses and see a connotation between now and your future job career.

Harmonie talked about some of the experiences she had in high school that allowed her to fully understand many of the basic math concepts that prepared her to master the more difficult concepts to help in later harder math courses. She story from high school like this:

In elementary school, I had a teacher who would apply songs to every single thing we learned in our math class starting with basic things like timetables and then moving to more difficult things like calculus and trigonometry concepts.

For Destiny and Tamera it's all about getting good grades and understanding the material so you can apply it to other harder concepts and assignments. Tamera puts it this way:

Um, success in math is more than just passing with an A or B grade it's about understanding the material and not just trying to retain it for that particular course or a test but actually, um, digging in, doing the practice problems, more than what the teacher gives you or is required in the textbook.

Success strategies

Several of the participants touted tutoring as a major strategy for doing in well in many of their math courses. In addition to significant importance was being able to have a positive relationship with the professors and visit with professors during office hours.

Other participants leaned on the support of other students who seemed to be mastering some of the challenging concepts. Jalyn talks about it in this manner:

For me going to tutoring and having a one-on-one with someone who can further explain the information to me that I wasn't able to get, like, within the classroom setting. So like really seeking tutoring support early.

Shelton states in this way:

Uh, for me, like, I've always been personally challenged with math so it's, one of those things I can't beat around the bush with. I have to work at it every moment and study daily so I can do well and pass my text and exams with A and B grades. In addition, a lot of mental math because in math classes here is different than in high where you were allowed to use a calculator. So, just practicing a lot of mental math. Studying over the review material before my test, uh, during the week, and going to tutorial sessions, you know, and getting the extra help I need is what's helped me prosper.

Williams forms it in this fashion:

I would say for me going to tutoring is the most important thing to do. Uh, that's basically all I could do or talk to my professor. If they're not willing to help you, then tutoring is the way to go because I've been to tutoring on many occasions and every time I've gone it's been extremely helpful.

Support systems

Among the many support systems, family and friends played a significant role in that process. In addition, tutoring services and other pre-college and in-college experiences lead to many of the success experienced by these participants. Harmonie expressed it this way:

Family because they always push me. You know, go get help, always get help. I was pretty shy growing up. I'm glad that I grew out of that but they were always, you know, on me, like, did you do this? Did you ask questions? Um, also friends. If you have, like, friends in similar situations

they'll push you to, uh, come on, let's go to tutoring, you know, let's go get help.

Destiny detailed it this way:

Um, yes all the time. Cause two brains are always going to be better than one. I may catch something that they don't see or vice versa. Or they can look at my mistakes and tell me what I need to do and things like that.

Technology Implementation

This area seemed to be a big deal for many of the participants and many had views that were similar and in many cases different but the consensus was having a physical textbook and manual homework was preferred over the online textbook and the online homework. The online homework they felt was too much work in terms of getting the right answers and truly understanding the material. They felt as though the notebook was too cumbersome trying to approach through an online platform. Here is what Toran said about it:

Okay, so with the textbook, I'm not a fan of the online textbook just because it's a huge transition to what I used to have during my middle and high school years. We always had the textbook during those times. We would refer to our textbook in class and pull them out. I mean we had it right in front of us at all times. And so I feel like that it being online, it kind of even makes the teachers not want to assign homework or even refer to certain page numbers within the textbook.

Harmonie talks about it in this manner:

Okay. So, everything being online it's, you know, it's modern nowadays, to have the technology. However, it's difficult for someone to manually go online and go to the book or go to the homework as opposed to it being right there in front of you. Um, the book being right there you don't have to, you know, type up, log in, pull up things when you can just, you know, open the book, read, learn and do all the required assignments.

Tamera reflects on it like this:

Um, we were a technology-based high school. We had like calculators that did everything. I mean, it could do a flip if you let it. Um, we mostly used the calculators for all our tests. We didn't do online assignments.

However, here at the university, we do a lot of Web-Assign. I think it's a blessing and a curse because it's like, um, I can do a problem on my own but if I get confused and need to know how to do the problem there is no step-by-step instruction as to how that should be accomplished. I don't prefer Web-Assign at all.

Christian B addressed it this way:

I feel like with the manual textbook it's much more to have the book in your possession because you can manually go through the book. You don't have to worry about the technology failing you at some point. Um, I would have to say technology at my secondary school, like, we were able to use promethium boards, um, which you can write on. So, you know, we'll go up and do, like, problems, if we didn't know, upon the board. The teachers would take the time to show us how to do the problems.

William articulates it in this fashion:

So the online book I have not looked at once. I just think it's useless. I'd rather have the physical textbook at my disposal. I'd rather carry a hard copy, because I mean it physically makes you look in the book, which I feel like is better. And then the online homework, I'd rather have paper homework as well because you have to be so precise with how you're entering answers and, like, if you mess up on a little comma or something of that nature you could end up getting the problem all wrong.

Pre-College Experience and Math Success

Early academic beginnings

In the early stages of starting elementary, middle, and secondary school there was a lot of pressure placed on the participants to succeed. Many had siblings who had set a path that they were required to follow. For many, they knew they were starting a path that would lead them to their intended major. Sean relayed his early academic beginnings like this:

Um, throughout middle school and high school, I've been going through, like STEM-oriented schools. So every class that I took each year had to do something around engineering, or something related. I always had a love for breaking and building cars. So I thought about engineering, and I just started thinking about pursuing a degree in mechanical engineering.

Toran was a student who began taking advanced placement courses at a very early age and was quite successful at passing the classes with A grades. Like Toran, Tamera also was placed in many advanced placement courses and did very well in those courses. Tamera attendance at the number one school in her state showed she had great academic potential.

Expectations of high school vs. university

Many of the participants had access to the use of calculators and technology while taking middle and high school math classes. They were provided with resources to help with learning difficult formulas and given equation sheets when taking exams or during testing but at the university setting this was taken away and provided challenges to many of the participants. Several of the participants relayed this dilemma below. Toran stated it this way:

In high school during a lot of my classes, we were almost forced to use our calculators without learning how to properly work many of the problems. However here in college, you are not allowed to use your calculators for tests or exams so this makes the learning process difficult at times. In high school, it was like, an acronym for everything. It was like some kind of easy way to remember all those difficult concepts. And with a lot of my teachers, you never had to remember formulas you were able to use an equation sheet. But here in college, you have to remember everything, which is not necessarily bad, but I feel like it should be a way in the classroom for students to have access to equation sheets for really long and cumbersome equations.

Christian S addressed it his way:

There were no computers for assignments but we always had access to use hand calculators. There was no working the problems out. Here, teachers teach you how to work the problem out without the calculator, which is very helpful. But coming from a school where you have those very powerful calculators the transition was very difficult.

Harmonie speaks to it in this fashion:

To a certain extent, there is a similarity between my high school and the way the university is flowing. We had access to calculators but we did not access to books most of what we did was through worksheets which were not very helpful because I feel we were not taught well using that approach. Sometime in the university setting, I feel as though sometimes with the online approach it's the same thing.

William talks about it in this manner:

I'd say in high school photo-math and Chegg was a really big help, but now, there are many problems where that technology will not work especially when trying to solve word problems. I've always been used to using a calculator in high school and for all my math classes, but here at the university I just have not been able to use a calculator in any of my math courses. This has been a challenging process for me to be successful.

Family expectations

The family expectation was pivotal for many of the participants and the families always set the bar high. Most of the parents were educated and understood the importance of providing that same type of education to their children. The participants felt the pressure to achieve and be just as good as their parents or strive for better. Several relayed their words below. Christian addressed it in this way:

In middle school and throughout high school my mom and dad always pushed me to excel. When completing assignments I was always timed to ensure I had speed and was checked to ensure I was correctly answering

the assigned problems. I could always remember it because I love the competition since I was the baby.

Jalyn spoke in this manner:

I am from a family of five. I have two siblings, a brother who's a freshman at Tuskegee, and a sister who is in the ninth grade. Both of my parents went to college. And they both earned their bachelor's degrees. My mom is in school for her master's in nursing now. They always stressed the importance to us of having a sound educational background.

Destiny relayed the information in this manner:

My mom has her master's degree and my aunts all have master's degrees as well. So it was kind of like, education has always been the key for us. My mother always stayed on me about my school work. I've never made a C, like average in a class ever. If I did make a C like I think I'll cry or my family just automatically will be pissed, disappointed basically. Um, I just come from a family of educators. Growing up it was always me and my mom. I'm the only child as well. My dad passed away when I was three. So it's always just been me and my mom. Like she's always been on me about education, make something out of myself. Basically, she did not want me to make any bad choices that would hamper my education.

Family Support

For many college students, family support is critical especially because of the financial constraints of students. Several of these participants were on scholarship however there were some who had not received scholarships and had to be supported by their families. Even though monies at times were difficult to come by they managed. Several students talk about their family support through hardships.

Sean describes it like this:

Um, for my family, mom went to college, and two of my sisters went to college. My dad did not, and my oldest sister tried but did not finish. Um, but, coming up I always kind of liked school. I always liked to stay in school. Mathematics was always my favorite course to do and was always

like a family kind of like a controversial thing to talk about. So that always helped out. I am on a general scholarship and must maintain at least a 3.3 GPA to keep my scholarship. Whenever I fall short on funds I reach out to my mother and father and they send the needed funds.

Harmonie puts it into words below:

I come from a very close-knit family so, you know, we depend on each other for literally everything. I grew up in a private school and then I went to a junior high, public school in the seventh grade. I've been in public schooling ever since. I have an older brother and a younger sister. We have always supported each other in all of our school work and academics. My family support means the world to me.

Tamera talks about it like this:

My mother has her doctorate in English, my dad started college but didn't finish. My mom works in the education field. She teaches at Harris-Stowe State University. She's been in every superintendent position starting as a vice principal and then a principal. My dad and mom have their own company and they have always supported us financially and in our academics. For then education is really important.

William addresses it this way as pressure on his shoulders:

Throughout my schooling, my family has been supportive of my entire academic journey. My father is a lawyer and so I have a lot of pressure on me to be successful. He supports me in all I do but still, there is the pressure that I have on my shoulders. Having two younger brothers I have to set a high standard to give them something to go after.

Extracurricular activities

Several participants were heavily involved in sports and other extracurricular activities. This provided outlets for these students who were additionally quite into their academics. Even though sports and extracurricular activities were important it never got in the way of their academic pursuits. A few students talk about their experience with sport and academics.

Christian S represented it in this manner:

Yes. I played football and track. The track was the sport I loved the most. I played football because my brothers played football and they all wanted me to play alongside them while they were playing.

Harmonie describes it like this:

I ran track for three years in high school. Um, I was a sprinter so I did several distance including the 100, 200, 4x2, 4x1, and sometimes long jump. In addition, I was in quite a few clubs including the Rotary Club and the debate club. I never allowed playing sports to interfere with what I considered more important.

Jalyn states it this way:

I played everything but basketball and track were my favorites. Here at the university, I'm on the track and field team participating in the long jump and high jump. I love university sports especially having the ability to interact with my pairs. For me being involved in sports helped with my math and other courses because weal ways had access to tutors in the Hankal Hall or the math and science building.

Destiny words it this way:

I played basketball year-round since seventh grade. It was very stressful trying to balance athletics and academics. I think it gave me some structure really because when I used to be in school, we'd have basketball practice maybe from 2:00. So after school, I would have to be on the floor to practice by maybe like 2:30 and so I would practice there, and then I would go do my homework when I got home. I always had that structure to do homework first when you get home. That's just how I was raised. I remember in middle school during the boy's practice I was set some time to complete homework and assignments before it was our time to practice.

Christian B states it this way:

Throughout my high school years, I played football. I also did track and field and the marching band. I was involved in this organization called DECA. So that was like a marketing club.

William relays the information in this way:

Well, I played basketball in high school, but I guess I'm not doing that here. But I'm in Honors College here so I'd say it's done nothing but help, just help me get in contact with the right people I need too and they send out tutoring information that also helps.

Self-belief and Motivation connected to Math Success

Believing in one's ability

Several of the study participants were consistent self-starters always wanting the best. These were students who excelled early in their academic career and always had an idea of where they wanted to be as adults. They were self-motivated and did not need to be push above their peer but to the contrary were pushed to success by others. It was some that same to be driven from the inside of these students who wanted better in life and to have a long-lasting career in their chosen profession. These student shares their views below.

Tamera puts it into words like this:

I've always been infatuated with science especially the teeth. From an early age, it is just something I always believed I would get into to pursue a career? I always admired the dentist growing up and I cling to people that help me grow as an individual and they always seem to take a liking to me and my interest in their profession. They were some of my early role models. They, um, push me to always strive for excellence especially in my academics knowing that getting into dental school will be a challenging process. So when taking difficult classes like math and biochemistry I always believed in my abilities that I would pass those classes with really good grades.

Christian B addressed it this way:

I feel like I could do anything. It does not matter how tough the classes I will always do my best to understand the material and pass all the tests and exams. Anything is possible because math is just something that, you know, people don't pick up on, but if you like excel in it and take the time

out to study it then, you're going to be successful at it. In addition, you will have success in other classes as well it's just a mindset and a discipline you have to take with you at all times and believe in yourself and your abilities.

Future math success

These students were faced with having to pass many difficult classes to become successful in every aspect of their academic life. However, for several of these students, it was something they had become accustomed to. It was something they had faced their whole life and nothing was going to stand in the way of them becoming college graduates and pursuing their intended professions. Even though the classes were difficult and hard to pass, they all felt as though not passing and fulfilling their dreams would be even more difficult. Some of their thoughts are expressed below.

Sean says it in this manner:

Having the opportunity to go to school with other students like myself was very important because I was able to see other students like myself having success in math and other STEM classes. Being an engineering student and having someone in my life who was an engineer was also beneficial because I got the chance to see what being a successful engineer is all about.

Harmonie puts it this way:

Being successful in this math class means I get closer to fulfilling my dream of becoming a physical therapist. I also see the ability to use what I learned in this and other math classes in my profession. For me personally, I want to restore people back to health and that involves telling people to continually and repeatedly do things over and over again, it also involves force and motions that may be difficult at first but with the counting concepts it will allow them to recover well.

Destiny addresses it in this fashion:

I do well in math even from high school and I have done well in all my math classes so far. I believe I will do in the ones I have remaining. However, the main reason I want to succeed in math and other STEM-related classes is because I want to become a dentist. So as a biology major doing the biology and math classes are just like peanut butter and jelly. It's better to have both rather than having one or the other.

Having balance with extracurricular activities

Being involved with extracurricular activities demands a lot of time and attention but not allowing the activities to take the place of academics was important to all of the participants. These students always remembered to put what's important at the forefront. When asked about if being concerned that extracurricular activities would become too much for students Toran states it this way:

It, it depends on the student because a lot of people get more caught up in the fun that they're having on the outside but definitely can be done with the good balance of, "I'm going to make sure this is done and I'm gonna make sure this is done." But a lot of people get lost until the fun or you know, the things they're doing outside of their coursework. So it's, it's about a balance and those students.

Harmonie states it this way:

It was not a problem for me. It didn't affect me learning because I knew when I was in school that was my main focus, and when I went home and I was done with everything related to school then that's when I could move on with my extracurricular activities.

Math success motivations

Being able to thrive in the absence of using technology was a must for all of these participants because they were not allowed to use calculators for their exams and test. This was something they had learned to use over their entire high school and middle school years so this was difficult. There were times when tutoring was needed but they

were skeptical to go because they always felt they were able to handle and difficult concepts. Overall the students wanted to be successful in their intended major and course in math and other related STEM courses and they were motivated to do just that. Several students below relay their thoughts.

Toran places words to it in this manner:

Being successful in math and being motivated has helped to shape my mind and have me confident about being able to pass all of my math and science classes. For me, nothing is really difficult you just have to decide that you are going to do whatever it takes to succeed. You don't say you can't do it but rather you say I must find a way to get it done. Like a lot of the stuff we did was depending on the calculator. So while all three of my math classes that I've had have had, had to take no calculator were allowed so it kind of expands your brain a little bit 'cause you have to switch from, "Oh let me pull out the calculator to let me get a piece of paper and a pencil.

Sean talks about it like this:

At first, I did not want to go to tutoring because I always felt as though I can handle all of the tough classes and work but of late I have gone to several tutoring sessions and has found it to be really good they go over the concepts slowly and they want you to understand the material. I still do most of my work to this day by myself but If I get stuck on something I go to tutoring because I don't want to fall behind in my classes. After this class, I want to continue to do well in my other math classes and my major classes, and whatever I need to do to be better prepared is what I'm willing to do.

Shelton speaks to it in this fashion:

For me C's are not acceptable, you know, B's are okay but my goal is to always seek after an A grade. You know I and my siblings have always been on the Principal's List or Honors Roll with perfect attendance, and good grades. I've always been personally challenged with math so it's, it's one of those things I can't, beat around the bush with. I have to work at it often to do well. I have to study like almost every day constantly, so I can

get an A in a test or, or a pass an assignment. I don't want to be the one who fails any classes so I must work hard to ensure that it doesn't happen but I continually move towards being finish with my degree.

Promoting a lasting legacy for African American Female STEM success

Many of the female students were consistent with wanting to empower other young women to pursue careers that maybe were never thought off. They also wanted to give back to the community and be a source of help for those who found themselves in very dire circumstances. Being blessed to have great parents this seemed in some ways to be a thank you for all their parents had vested in them. Several of the female participants share their views. Jalyn addresses the issue in this manner:

I chose my major because I loved working in the medical academy at my High School. In my senior year, we did a clinical rotation at different hospitals where I had the opportunity to see if this was something I wanted to do. I already somewhat knew I was into physical therapy and it was going to be something I did on a long term career path. I felt as though this would be my opportunity to give back to my community for all the support I had received throughout my growing up years.

Tamera describes it in this way:

I like to help people and I believe being in the dental line of work would provide me with a unique way to give back to others who are less fortunate. I've always had access to many types of medical treatments and dental treatments and I know a lot of people don't have that same opportunity so I want to be in a position to give to those who are in dire circumstances. I never struggled so I don't know how it feels to not have dental insurance. I don't know how it feels to be without and thanks to my parents I will have this opportunity to make a difference for many others. I love working with teeth and I believe with good oral health you can have great overall health.

The HBCU Experience and Math Success

HBCU effect on math success

Attending an HBCU for many of these participants was a bucket list so to speak. They felt as though the environment would both be nurturing and tough but also allow them to have individual growth and success as they progress through courses. Resources were in place to assist when and wherever necessary. There was no belittling and there was always a sense of community and belonging. Talking to professors was easy and in some ways, they knew the struggles you faced to get to this level. Even though at times there was competition it was good competition because the goal was to get everyone at an acceptable level. Below some of the students address their view of the experience. Sean puts it into words in this fashion:

I would say attending an HBCU the main important thing for me was the environment. I would say the professors here want to help and point you in the right direction if you need additional help. Even if you're doing well they always stress the importance of keeping up with the work and being up to date with all homework and assignments. So they always push you to get more help, such as taking advantage of tutoring or talking with other students who are doing well understanding the concepts. They are always pushing you forward to do better.

Destiny speaks to it in this manner:

One of the main reasons why I chose to attend an HBCU institution is due to the fact of the historical perspective. I feel like the class sizes would be smaller and the professors would relate to you better as an African American student. It also would be good for me to see students who are like me excelling in their courses; this would give me a boost of confidence knowing that students with probably some of my same struggles are doing well in their classes. The environment seems to be pleasant and you can build a sense of pride before going into the real world. You also get to build lifelong friendships over the course of the

years and you feel as though they have your back. Many resources are in place to help ensure you have every opportunity to be successful

Christian B puts it into words in this way:

The experience of attending an HBCU is just different. The math classes here are hard but I never really feel like I can't do the work or pass the class because of the many vital resources available to help with my success. When I talk to my friends who go to, PWI they say their classes tend to be hard because they expect you to know everything even if you did not cover those things in high school. Also, I'd have to say the professors are willing to help. In addition, the university and the different departments provide resources to ensure that all the students have a chance to succeed. The HBCU feels like a space where both nurturing and growth can take place without hindering the student. Even though they are willing to help they also require you to put forth effort in achieving the results you desire. The school spirit is great and I feel like I'm a part of this larger community.

Help from professors from HBCU institutions

Jalyn voiced it in this fashion:

At the HBCU definitely, you can build upon a close-knit relationship with the professors. They are always willing to help and you can freely attend the office hour sessions regularly. With the class sizes being really small they get to know you by name and not just a number.

Christian B puts it this way:

For sure I would say the teacher and professors are relatable on a one to one personal level. They are often wanting and willing to help in and out of the class. Many of the professors assist with tutoring so you can get the one on one help without disturbing the class or looking stupid in front of the other students.

William addresses it in this manner:

I would say the classes are smaller here at the HBCU so it gives me more one-on-one time with my professors. So I'd say it's helped for sure 'cause I can just talk one-on-one with my professors' right after class if I need to.

If they are not available after class you can go to their office or tutoring because some of the professors also tutor one on one.

Importance of the HBCU connections to math success

Attending an HBCU institution provided a solid foundation for these beginning African American students. They were taught by many African American professors. Their interaction was similar to the other students, the class size was not too large and they felt the teachers were more approachable and had a better understanding of the struggles and challenges they face. Many of the participants relayed their views about the HBCU connection to help with math success. Sean speaks to it in this manner:

I would say going here has been a really good experience because of the environment of everyone wanting to help you succeed. They always seem to push you to do better and not just settle for anything less. Even if you're doing well they always advise you to continually seek help it's never too much to ask for more help. The tutoring is really good because it helps you to better understand the concepts that you may struggle with within the class.

Christian S voiced his views in this fashion:

Um, attending an HBCU contributed to my success by allowing me to see the experiences of others who have similar backgrounds as I do. I don't feel alone. So getting through a hard major like civil engineering is just something I believe I could accomplish being here at this university.

Jalyn shares his views in this manner:

For one thing, I consider it an honor to be attending an HBCU institution. I've had the privilege of growing up and being able to go to various primarily white pre-college schools. I'm from Alabama, so you know everybody thinks of Auburn or Alabama; I never thought about attending those two institutions for me it was always going to be an HBCU. HBCU always seems to have a close-knit, family structure, and the support there is second to none. Also with our class ratio-it makes it easier to develop a

relationship with the teacher. I'm not just a number among the teachers; they want to know my name.

Destiny expresses it this way:

One of the main reasons why I chose to attend an HBCU is because of its historical perspectives. I felt I would get a better understanding of my history without false information. And I felt like I would get a better understanding of my math courses attending HBCU because of the class size. I know at a PWI, the class sizes may be like 60 to 100 people. You can't build a relationship with your professors, really I feel like you're just a number, um at a PWI. I feel as if at HBCU, you can form a relationship with your teachers. You can come to the office hours. You can build a relationship with them, and so they can focus on what the student needs.

Women in STEM and Math Success

Access to STEM for African American females

When attending many of the classes I noticed there were almost equal amounts of men and women in the math classes. I also noticed the women were sitting closer to the front of the class compared to the men. This was significant because even though society may view women differently than they do men in the STEM fields the women in the classes seem to have not let that deter them from pursuing their intended major. Several of the female participants speak to this below.

Harmonie states it like this:

I feel, like, it's wonderful when women can have the same success as men in the STEM field. I feel, like, men and women are equal and are both capable of achieving great things in life. Gender shouldn't differentiate whether someone can be successful or not in a STEM field. There are times when the access of females may be limited because of the views society place on women. I have had many STEM teaches who inspire m to get involved with STEM. Because of these teachers, I'm now pursuing a STEM major.

Jalyn talks about it in this fashion:

I think when people ideally think of a STEM major they think of it being a major predominately where there a lot of males and that may be true, but I believe the STEM field is a place where women can thrive as well. When thinking about women in the STEM field I'm reminded of the movie hidden figure. In the movie, the women were critical in the success of NASA. Females can do the same thing as males even with a lot of uneven expectations.

African American women in STEM as role models

Participants consistently talk about the impact that African American female STEM teachers provided to them in their struggles. Many classes were equally filled with African American female and male students who were pursuing STEM degrees and on many occasions, the females provided more of the support to other students who were struggling. Several students talk about the experiences below.

Jalyn states it this way:

During many of my middle and high school classes, most of my STEM classes were taught by African American females and this inspired me to want to pursue a STEM career. When I got to the university setting I was surprised to see how many STEM classes were taught by females who were African American. In some ways I consider them to be role models because I always feel as though I can approach them and ask them questions relating to my STEM classes but also the journey they took to bring them to this point.

Shelton notes it in this fashion:

I feel like a lot of females I have in my class are just as smart as or even smarter than some of the males in the class. So, you know, they can learn the material just as well as we can. To me, there is no difference. In fact, many of the female students in my class assist me with a lot of my homework assignments and I often look to them for their help and guidance.

Tamera addresses it in this manner:

Being I many different organizations and seeing females as the head is inspiring. For me in the dental area is stressful and sometimes gets difficult and many students drop out of the program for various reasons. I think that you have to mentally prepare yourself for because it's not easy. I think, in my years of college, this is probably my hardest semester but because of my peers and many of the African American female STEM teachers, it has helped me to persist. They have mentored me and have become great role models.

Society's view of where women should be

Several of the participants had varied views of how society limits women to certain roles and often neglects the talents that reside in so many African American women in the STEM field. Toran and William both believed that society's limitations on women is wrong and believes women and men can have similar success in the STEM fields. Toran's view was this:

Female African Americans can definitely do STEM. Um, but I feel like society often misrepresents the place of a woman in general. Even if you look at society, they kind of tell you that women can be nurses, and women can be caretakers, and women can be doing these different things. And it's like you have to have those people who are willing to go back and tell them like, "I am a STEM major who is very good at what we do as females.

William had a similar view and relayed it in this manner:

I believe African American women in STEM can. I believe that whatever they put their mind to they can achieve, so if you're a female in the STEM field, I think you have just as much opportunity as a male, so I still feel like that females, even with the gaps that may exist have the opportunity to be successful.

Sean and Jalyn noticed the disparities from a societal perspective that placed biased on who gets hired in the STEM fields and relayed how it can be a male-dominated area even though females are capable of being successful. Sean states it in this fashion:

Um, I would just say, basically, kind of like when you're applying at jobs, there are more male-based bosses, and everything like that and they tend to be more averse to hiring male counterparts. They really, kind of if they see that you're a female, not saying all people, but if they see like you're a female, they kind of just have doubts in your abilities from the very onset.

Jalyn views it in this manner:

Um, I think when people ideally think of, like, a STEM major, I think they think it is more a field where males are usually more successful. Um, so, if a female comes through there succeeding in it I believe it's amazing. Um, it's just a positive impact, you know. Females can do the same thing as males, being in many ways held to a different standard.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATIONS

Summary of the study

To finalize this dissertation study, this chapter will provide another look into the problem statement and the methods used to undertake the study. Included in this chapter will be the summary of the study, limitations of the study, and an overview of the results which discusses the findings which are organized around five themes: (1) Perceptions of Math Success (2) Pre-College Experience and Math Success (3) Self-belief and Motivation connected to Math Success (4) The HBCU Experience and Math Success and (5) Women in STEM and Math Success. Within these five themes, we begin to understand some of the pre-college experiences, in college experiences, and some of the aspirations for post-college experiences that produce driving attitudes for the success of African American STEM students finding and experiencing success in university-level mathematics courses. This chapter ends with conclusions and implications along with recommendations for future research.

The purpose of this study through conversations with African American students in Science Technology Engineering and Mathematics (STEM) disciplines, who have achieved success in mathematics, was to explore and determine the perceptions these students had about how contributing factors benefited their overall success in mathematics and strategies they employed to assist with that mathematical success. In addition, the issue of African American female students from those STEM disciplines

was addressed to understand how perceptions and notions may have tried to prevent them from having similar success as African American males in obtaining STEM degrees and being successful in university-level mathematics courses. Lastly, this study allowed the participants to provide rich, non-numeric data, and qualitative information to be captured providing a naturalistic approach to the involved subject matter and allowing meaning to come from the study participants' perspectives while giving voice to a minority group whose voices are seldom heard Chen (2005).

The overarching research question for this study was: What are the perceived and contributing factors of African American students that may contribute to their success in university-level mathematics courses and what strategies have they used to achieve that success. In addition to answering the overarching question several further questions were developed:

- 1) How do African American STEM students perceive success in mathematics at a Historically Black College and University (HBCU)?
- 2) How have experiences of African American STEM students (e.g., past and present) contributed to their success in mathematics?
- 3) To what extent has being successful in mathematics contributed to overall self-efficacy and motivation of African American mathematics students?
- 4) How has attendance at an HBCU provided opportunities to meet the needs of African American students in STEM related discipline?

- 5) Is there a perception of African American STEM female students that gender plays a role in their involvement in STEM related disciplines and success in undergraduate entry-level mathematics courses?

Overview of the Problem Statement

Low passing rates among first-year university mathematics courses especially College Algebra and Pre-Calculus presented a major concern and challenge for university stakeholders. According to research African American students are usually lagging behind other ethnic groups when it comes to preparation for post-secondary success (May & Chubin, 2003). When comparing African American males and females, female students are usually less likely to take math classes beyond Calculus I or pursue a STEM-related major (May & Chubin, 2003). This circumstance in many instances prevents female African American students from perusing STEM careers and usually contributed to the low representation of African American females in STEM-related fields.

Furthermore, when examining classes with only African American students enrolled, the failure rates for traditional class settings (classes with formal teaching instruction) are 55% higher than nontraditional class settings (classes combined with both formal instruction and technology implementation) (Freeman et al, 2014).

Summary of the Methodology

The method employed to undertake this study used an interview interpretative style approach to gather information from African American Science Technology Engineering and Mathematics (STEM) students who have successfully passed college algebra and/or pre-calculus with a grade of A or B. The investigation sought to gain

understanding from participants' experiences as to what constituted success in mathematics and how the identified factors may have contributed to that success.

To address this research study, a qualitative research design was employed using a Phenomenological research analysis. This allows for valuable data to be gathered from research participants regarding their experiences relating to a concept. Blue University (pseudonym) was used for the setting of this study. Blue University is located in the southeastern region of the United States with a student population of under 8,000 students consisting of both undergraduate and graduate students. The undergraduate student-faculty ratio is 17 to 1. The population for this study was STEM students enrolled in Pre-Calculus II and Calculus I who had completed college algebra and/or pre-calculus with an A or B grade.

51 students were interested in participating in the study and filled out the initial questionnaire. Of the 51 students, only 29 met the criteria for the study. Of the 29 students who met the criteria, ten were randomly selected. The ten selected were called and informed about their selection and times and dates were set up to start the interview process. The data was collected through face to face interviews using a standardized series of open-ended questions derived from the survey questionnaire. Questions were randomly selected and answer responses were further questioned for additional clarity. Notes were taken and voice recording took place for later review and analysis.

Data were collected through two methods, (a) a demographics questionnaire and, (b) a specifically design open-ended survey questionnaire. Questions on the demographic survey included age, gender, parental income status, and first-generation identity. In

addition, students responded to a series of open-ended questions from the survey questionnaire. Qualitatively, question responses from the open-ended survey questionnaire were used to glean information to provide context and structure from the student responses and seek answers to the study questions.

With the proposal and IRB approval, a permission letter was sent to the mathematics department head to provide information about the study and seek permission to communicate the study with individual classes. Upon approval, the researcher met with selected faculty members to request permission to speak with their respective classes. The interview process took between 30 to 45 minutes to complete. After an initial interview, a second member checking interview was conducted to ensure participants were in agreement with the interpretation of the information gathered.

Once interviews were completed the data was sent off to be transcribed. Once returned notes were added to ensure completeness of thoughts; organized with study narratives and cross analysis then finally coded to find patterns and themes that sought to address the study questions and find context within the existing literature. The data collected was analyzed with the help of Dedoose, which is a qualitative software that helps with the process of qualitative research analysis. To ensure complete privacy pseudonyms were used for all the participants. The Creswell Data Analysis spiral approach was used which seeks to provide understanding at various levels of the data analyzing process. All information collected was kept secure through password protection and lockbox facility to provide confidentiality to the study participants.

Limitations

This study considered only STEM students enrolled at Blue University who had passed a freshmen level algebra or pre-calculus course with an A or B grade and were currently enrolled in an upper-level mathematics course beyond Pre-Calculus I. Due to the design of this study, the findings would not be generalizable to other non-STEM majors. This research study was significant to the researcher because similar circumstances were encountered by the researcher being a first-generation student who was a low-income student in addition to being a STEM major. The researcher had also worked with many African American STEM students in mathematics and had thought about what constitutes success from their perception and perspectives and what allowed one group to be successful while the other group struggled. As a point of emphasis, the researcher “bracketed” as much as possible to relay the lived experience of the participants. Bracketing allowed the researcher to be unbiased in the assessment of the data collecting and interpreting process (Carpenter, 2007).

Overview of the Results

Within this study, the findings reveal that many of the participants perceived mathematical success to be more than a single element. For many of these participants, mathematical success consisted of at least several things including; (1) receiving good grades, (2) overcoming math obstacles, (3) having great institutional and family support, (4) being balance through academics and extracurricular activities, (5) being able to fully understand the math content, (6) receiving help from professors and peers, (7) having access to tutoring, (8) liking mathematics (9) seeing a linking to a future profession, (10)

seeing their role as students in mathematics as important and (11) eliminating gender biases. The research questions with interpretations and literature to support the findings are now relayed.

Research Question One (Perception of Math Success)

How do African American STEM students perceive success in mathematics at a Historically Black College and University (HBCU)?

For many of the participants, perceptions of math success were understood to be focused on several important aspects; (1) maintaining academic excellence, (2) implementing technology with non-technology based learning, (3) overcoming difficult math obstacles and math instructors, (4) understanding difficult math content, (5) a natural liking of mathematics and (5) receiving good grades was often heard among the participants. For the most part, the students realized that in spite of any difficulties or problems the ultimate goal was to pass the class with a decent grade. This stance is consistent with findings from other studies such as M. L. Johnson (1989) who suggest that obtaining good grades was the most important factor for being a success in university-level mathematics courses. Also, when Powell-Mikle (2001) examined the participants during their study they concluded the majority of their study participants found a liking of mathematics to be positively associated with good feelings of math interaction and greater outcome of math success in university settings. Many of the participants of this study did not think they were smarter than other students but they had the ability to apply themselves greater than their peers to obtain the success of math content acquisition and overall mathematical success even through difficulties and trying

obstacles. Powell-Mikle (2001) would consider these types of students very successful having overcome continued deterrence in their pursuit of academic stability and success. Seeing the need to fully understand math content and acquisition and the relevance of the math to future career goals made learning math and applying the math to other related classes made the math outcome a major stepping stone to the pertinent mathematical success perception. This is consistent with (D. Martin, 2000; Powell-Mikle, 2001; Thompson & Lewis, 2005) who believes that when clear achievement goals are known and laser-focused, African American students tend to be motivated in the pursuit of having academic and mathematical success.

The study participants had great support throughout their pre-college years in elementary school up to high school this support provided a means for many of them to have a beginning love of what and how math success is tied into parental and community support. This type of support and rigor for pre-college experiences leading to mathematical success is consistent with findings from several authors who suggest that when African American students have pre-college rigor and vital support systems they will achieve greater academics and mathematical success (Gutierrez, 2000; Moody, 2004; Powell-Mikle, 2001; Thompson & Lewis, 2005).

Research Question Two (Pre-College Experiences and Math Success)

How have experiences of African American STEM students (e.g., past and present) contributed to their success in mathematics?

The participants interviewed were from a mixture of both single and two-parent families. In addition, some families were financially secure and then others were making

it but had some financial struggles. Participants came from large families where there were 17 siblings and some came from families where they were the only child or had a few siblings. Under all circumstances, the support of the family was a critical component in the trajectory of whether these particular study participants would succeed or become a part of the statistic of those who were unsuccessful in the math courses. This was a major component of the social and cultural capital of these participants which are ways and means to provide educational resources to support academic integrity and to ensure that culture is present and supported in academic settings. Surprisingly, all parents of the participants that lived in their homes were educated. There were several associate degrees, bachelor's degrees, master's degrees, and even doctoral degrees. Some of the participant's parents were even business owners. The knowledge of knowing that their parents were college graduates encouraged these students to want the same things for themselves and provide their parents with joy for doing such. This is consistent with several studies (Bush, 2004; Strutchens, 1993) especially research in higher education which supports the fact that when parents are well educated it is a significant indicator of greater college grade point averages, persistence in academic endeavors, and overall success in university settings and major coursework (Bush 2004). In addition, depending on parent's educational attainment, the interaction with university subjects, especially mathematical content, can be enhanced and allow African American students to perform better in mathematics classroom settings (Strutchens,1993). Also, Oaks (1990) suggest that when parents are educated they have the opportunity to place themselves in a different socioeconomic status. This gives them the ability to provide additional

educational resources to their children which provides the children with a myriad of educational opportunities. This gives the children additional opportunities to be prepared to excel academically within a university environment. Furthermore, Nora and Cabrera (1996) suggested that “parental encouragement and support was found to exert a positive effect on the integration of students to college, on their academic and intellectual development, and on their academic performance and commitments” (Nora & Cabrera, 1996, p. 140). Along with parental and community support, early educational experiences were also significant for it allowed these participants to realize a sense of worth and importance of early mathematical foundations. They also seemed to have caring teachers who were able to identify with their backgrounds and be valuable role models for them in uncertain times. These teachers challenged the students and provided an early independent thinking process to help the students to mature academically.

Research Question Three (Self-belief and Motivation connected to Math Success)

To what extent has being successful in mathematics contributed to overall self-efficacy and motivation of African American mathematics students?

These participants all seemed to be conscious of what was required for academic excellence. They credited their parents and early schooling as important factors in their academic development but realize at some point along the journey that they needed to be responsible for their academic progress and success in their mathematical courses. Even though they had difficult experiences in mathematics while growing up they never doubted their ability to succeed and become really good at understanding math and being able to excel. According to Hackett and Betz (1989), mathematical self-efficacy can be

thought of as a person's ability to both succeed in math and relay math content in a manner to provide assistance towards anyone struggling. It has also been shown to be a reliable indicator of a person's ability to be successful in both mathematical achievement and mathematical performance. When understanding the participant's mathematical abilities there seemed to be no unity between mathematical abilities and the participants' attitudes. From Hackett & Betz (1989) study they said it in this manner: "examinations of a person's mathematical ability are unique from other measures of attitudes towards mathematics in that mathematics self-efficacy is a situational or problem-specific assessment of an individual's confidence in her or his ability to successfully perform or accomplish a particular task or problem" (Hackett & Betz, 1989, p. 262). Varying degrees of self-efficacy were detailed by many of the participants and were depending on different and unique circumstances. Obstacles and tasks in the mathematics courses were problematic at times but this did not prevent the students from being fully confident in their abilities to face and surpass the challenges. For the most part, their overall level of motivation and self-efficacy was correlated to their ability to be successful in mathematics courses and create a stable sense of mathematical identity.

Research Question Four (The HBCU Experience and Math Success)

How has attendance at an HBCU provided opportunities to meet the needs of African American students in STEM related discipline?

Through mostly positive interactions with professors and peers along with the nurturing environment of the HBCU, participants were able to thrive even when faced with difficult demands such as passing courses, meeting professor expectations, and

managing the rigorous demands of no longer being a high school student but rather a university student who is adjusting to a new environment. For these participants having a good relationship with professors was a top priority because they felt this provided a means for allowing the professors to know who they were and also their seriousness in regards to passing the classes. What most participants noted about these interactions was the authenticity of the professors in really wanting to know them and provide them with the most realistic options to allow them to achieve not just mathematical success but overall university success leading to positive future career outcomes. Also of importance, was the uniqueness of the HBCU experience because it allowed them to see other African American role models who had achieved the success they were pursuing. This finding is consistent with other research by Pascarella & Terenzini (1978) and Terenzini & Pascarella (1980) which suggest that positive interactions and relations with professors led to a greater degree of academics success and persistence among university students. In addition, Powell-Mikle (2001) noted that when students are seeking help and find professors who are willing and available at regular times these were viewed as professors who cared and had the best interest of the student and were able to make difficult math content more understandable.

Another key component to the HBCU experience was the university itself which ensured that resources were in place to meet the needs of these students in areas where additional support was needed whereby they could have success in their math and other related courses. There were also clubs and organizations that supported the mission of universities was is to educate in preparation for post-college careers. This finding is

consistent with findings by Tinto (1998) who suggested that students who are goal-focused usually are influenced by interactions within the university both at the social and academic levels.

Participants alluded to the fact that having peers with similar backgrounds and similar stories was also a critical component of the HBCU experience. For many having peers who were similar provided consistent motivation for pursuing and continuing in their intended major along with removing the feeling of isolation and fear and providing a roadmap for academic success. Not only was peer interaction important for the above mentioned components, but it was also important for tutoring and study sessions which were able to be areas for deeper bonding as well. This is consistent with finding by Cobb (1994), Cobb & Yackel (1995), and Vygotsky (1994) who suggest a sociocultural perspective to comprehending mathematics is for students to construct mathematical competency and understanding by in-depth interaction with peers who are proficient in their mathematical ability. It is also in a line with Brew (2002) who found that having different perspectives through peer interaction provided the necessary mathematical support to increase mathematical content knowledge and application. This interaction also forged healthy competition among the learners' especially African American learners and provided encouragement in times when emotional support was needed and this was consistent with research by Martin (2000).

Research Question five (Women in STEM and Math Success)

Is there a perception of African American STEM female students that gender plays a role in their involvement in STEM related disciplines and success in undergraduate entry-level mathematics courses?

Surprisingly, all of the females in this study had encountered some form of gender-biased during their early schooling years and slight among their college years however none of the biases swayed the participants from continuing to pursue STEM-related disciplines and having success in STEM-related courses, especially mathematics. Attending an HBCU institution they felt as though their African American male counterparts saw them as equal and did not believe that they had a lesser ability to perform similarly as themselves. They would often study together and in many instances, the female African American STEM students would lead the study and tutoring sessions. The females in this study believed this feeling of identity and self-security was mainly because of the HBCU environment. This is consistent with Astin's (1993) study which suggests that students bring to university characteristics and experiences from past experiences that intermingle with the university environment to produce consistent skill sets and new abilities. The women in this study had a very secure sense of ability and purpose to drive their matriculation.

In addition, Padilla's (1999-2000) study suggests that universities can be viewed as a "geography of barriers" (Padilla, 1999-2000, p.136). The combination of barriers and student abilities to overcome these barriers for many of the women in the study was of paramount and significant importance because it gave credence to what they were able to

accomplish. Many of the females in this study had a strong academic and mathematical background from early school beginnings and whether they realized it or not this was helpful and necessary for success in college-level classes especially mathematics. This was consistent with what both Astin (1993) and Pascarella & Terenzini (2005). The female students in this study had a practical consideration of self-ability and knew additionally when necessary to seek help when necessary. They considered seeking help, not a weakness or thinking it's only something females did but rather a mechanism to seek out clarity to master their mathematical courses and content. Building strong relationships with female STEM teachers and peers through mentorship programs was also a significant contribution in allowing these female students to become confident in their roles as future STEM majors and having an impact in future career fields predominately occupied by males.

Conclusions and Implications

During this study, the participants provided several factors and contributors that allowed them to be successful in their mathematical experiences throughout university-level mathematics courses as well as mathematics courses taken at the pre-college level. The important factors were in some ways tied to family support systems, pre-educational experiences, community support, in-college support and experiences, and also personal motivational factors. To delve deeper into the understanding of this particular group of successful African American students in STEM disciplines additional research is still needed however the study findings are consistent with previous research on this selected phenomenon of mathematical success. The findings of this study suggest that the pre-

college experiences played the biggest role in the development of these participants' mathematical success development. Parents, having obtained regular and advanced degrees provided some of the necessary motivation of having to succeed, and knowing the eye of the community and younger siblings were looking made the journey to success even much more rewarding. Not to be outdone by the pre-college experiences the in college experiences were just as important especially having a faculty support system, a peer support system, and a university support system. In addition to having the pre-college and in college support, these participants relied on a post-college career outlook to ensure that even in difficult circumstances along the process they were going to not just be successful in the mathematics area but in all areas that would lead to a secure future outcome. Their self-efficacy and motivation may have been primed through initial experiences and they continued because these participants wanted to finish strong what they started, seeking to become educated African American STEM students.

This study can have implications for several stakeholders inclusive of (1) mathematics professors, (2) higher educational administrators, (3) higher educational housing staff, (4) higher educational IT staff and, (5) higher educational athletics. This study can help mathematics professors better understand the dynamics of the African American STEM student populations and connect instruction and content which relevant learning experiences. Secondly, higher education administrators can benefit from this study by ensuring programs and policies are in alignment to assist disadvantaged students, especially those of the African American student population in STEM disciplines. Thirdly, higher educational housing staff can use this study to help African

American STEM students to transition effectively between secondary and post-secondary institutions while gaining independence. Fourthly, higher educational IT staff can use this study to ensure infrastructure is up to date to allow African American STEM students to use the technology and technology platforms to produce optimal level work. Finally, higher educational athletics personnel can use this study to ensure that academics and athletics work together to produce a student who can traverse both the academic and athletic commitments.

Recommendations for Future Research

The research in this area is limited and even though this study adds to the limited literature more longitudinal studies should be conducted with some of the major relevant themes. Also, adding a quantitative component together with the qualitative analysis to capture a larger perspective to better understand the experiences of successful African American STEM students especially those in the critical area of mathematics.

This study focused only on African American students. Additional studies could be conducted to determine if similar findings could be found among other ethnic groups. In addition, a comparative analytical study could be conducted with both African American students and other ethnic groups within the same study to determine if similar findings would be observed. Similar studies could be conducted at PWIs to determine if African American STEM students within that community would have similar experiences like those at Blue University.

Further research could be conducted to determine not only how African American STEM students achieve mathematical success but what factors prevent this same group

from remaining in STEM fields and having similar success as those who remain in the STEM majors. Another area to consider would be a comparative study to determine if non-STEM majors at HBCUs would share similar findings as found in this study with STEM majors.

REFERENCES

- Abouserie, R. (1995). Self-esteem and achievement motivation as determinants of students' approaches to studying. *Studies in higher education*, 20, 19-26.
- Ackerman, P. L., Kanfer, R., & Beier, M. E. (2013). Trait Complex, Cognitive Ability, and Domain Knowledge Predictors of Baccalaureate Success, STEM Persistence, and Gender Differences. *Journal of Educational Psychology*, 105(3), 911–927.
- Advisory Committee on Student Financial Assistance. *Access Denied: Restoring the Nation's Commitment to Equal Educational Opportunity*. Washington, D.C.: Advisory Committee on Student Financial Assistance, 2001.
- Alfaro, F. (2008). *The relationship of teacher perceptions of the impact of technology integration on Texas assessment of knowledge and skills (TAKS) scores of 9th to 11th grade students at Alamo Heights Independent school district, San Antonio, Texas* (Doctoral dissertation). Retrieved from ProQuest Dissertations and Theses database. (AAT 3321629)
- Anderson, B. (1990). Minorities and Mathematics: The new frontier and the challenge of the nineties. *Journal of Negro Education*, 59(3), 260- 272.
- Anderson D, and Kim D, (2006) Increasing the success of minority students in science and technology. Washington, DC: American Council on Education.
- Armentrout, W. D. (1978). Neglected values in higher education: Needed reorganization in curricular and extra-curricular activities to provide significant experiences. *The Journal of Higher Education*, 50(4), 17-20.

- Arnold, A. (1999). *Retention and persistence in postsecondary education – A summation of the research studies*. Texas Guaranteed Student Loan Corporation.
- Ashwin, P. (2006). *Changing higher education*. Hoboken: Routledge.
- Astin, A. (2001). *What matters in college?* San Francisco: Jossey-Bass.
- Astin, A. W. (1993a). What Matters in College. *Liberal Education*, 79(4): 4-15.
- Astin, A.W. (1993). *What matters in college: Four critical years revisited*. San Francisco: Jossey-Bass.
- Bailey, T. R., and Alfonso, M. (2005). *Paths to Persistence: An Analysis of Research on Program Effectiveness at Community Colleges*. New Agenda Series, 6(1). Indianapolis, IN: Lumina Foundation for Education.
- Bandura, A. (2001). Social cognitive theory: an agentic perspective. *Annual review of psychology*, 52, 1-26.
- Banta, T. (2001). Learning communities illustrate principles of good practice. *Assessment Update*, 13(4). Retrieved August 15, 2016, from Academic Search Premier.
- Barovick, H., and Baron, M. (2001). *Indiana University: A Web of Friendly Interest Groups Makes This Big Research Institution Feel Less Intimidating*. Time, September 10.
<http://www.time.com/time/archive/preview/0,10987,1000722,00.html> (accessed January 2, 2017).
- Bean, J. P. (1983). The Application of a Model of Turnover in Work Organizations to the Student Attrition Process. *The Review of Higher Education*, 6(2): 129-148.

- Berkeley-Jones, C. S. (2012). *A study of the relationship between levels of technology implementation (LoTi) and student performance on Texas Assessment of Knowledge and Skills (TAKS) Scores* (Doctoral dissertation). ProQuest Dissertations and Theses database. (UMI No. 3532058)
- Bloom, H. (2006). The advisor's guide to serving simulation junkies. *Campus Activities Programming*, 37(7), 17-20.
- Brew, C. R. (2002, April). *Peer-interactions in the adult mathematics classroom and the flow-on effect to the children of women returning to study mathematics*. Paper presented at the annual meeting of the American Educational Research Association, New Orleans, LA.
- Bush, E.C. (2004). Dying on the vine: A look at African American student achievement in California community colleges (Doctoral dissertation, The Claremont Graduate University, 2004). *Dissertation Abstracts International*, 64, 4326.
- Cabrera, A. F., Casteneda, M. B., Nora, A., and Hengstler, D. (1992). The Convergence Between Two Theories of College Persistence. *Journal of Higher Education*, 63(2): 143-164.
- Carpenter, D. R. (2007). Phenomenology as method. In H. J. Streubert & D. R. Carpenter (Eds.), *Qualitative research in nursing: Advancing the humanistic imperative* (pp. 75- 99). Philadelphia, PA: Lippincott.
- Cavaye, A. L. M. (1996). Case Study Research: A Multi-Faceted Research Approach For IS. *Information Systems Journal*, 6(3), 227-242.

- Chamorro-Premusic, T., & Furnham, A. (2003). Personality predicts academic performance: evidence from two longitudinal samples. *Journal of research in personality, 37*, 319-338.
- Chen, X. (2005). *First generation students in postsecondary education – A Look at their transcripts (Exchange Summary)*. National Center for Educational Statistics. Retrieved September 4, 2017, from nces.ed.gov/das/epubs
- Choy, S. P. (1999). College Access and Affordability. *Education Statistics Quarterly, 1*(2): 74-90.
- Chrispeels, J. H., and Rivero, E. (2001). Engaging Latino Families for Student Success: How Parent Education can Reshape Parents' Sense of Place in the Education of Their Children. *Peabody Journal of Education, 76*(2): 119-169.
- Christy, R. D., & L. Williamson, eds. *A Century of Service: Land-Grant Colleges and Their Universities, 1890-1990*. New Brunswick, N.J.: Transaction Publishers, 1992.
- Cobb, P. (1994). Where is the mind? Constructivist and sociocultural perspectives on mathematical development. *Educational Researcher, 23*(7), 13-20.
- Cobb, P., & Yackel, E. (1995, October). *Constructivist, emergent, and sociocultural perspectives in the context of developmental research*. Paper presented at the annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education, Columbus, OH.
- Creswell, J. W. (2007). *Qualitative Inquiry and Research Design: Choosing Among Five Traditions (2nd ed.)*. California, U.S.A: SAGE Publications.

- Creswell, J. W. (2012a). Educational Research: Planning, Conducting, and
*Evaluating Quantitative and Qualitative Research (4th ed., International edition
 ed.): PEARSON Publications.*
- Creswell, J.W. and Poth, C.N. (2018) *Qualitative Inquiry and Research Design Choosing
 among Five Approaches*. 4th Edition, SAGE Publications, Inc., Thousand Oaks.
*Evaluating Quantitative and Qualitative Research (4th ed., International edition
 ed.): PEARSON Publications.*
- De Raad, B., &Shouwenburg, H. (1996). Personality in Learning and education.
European journal of personality, 10, 303-336.
- Duckworth, A., Peterson, C., Matthews, M., & Kelly, D. (2007). Grit: perseverance and
 passion for long-term goals. *Journal of personality and social psychology, 92,*
1087-1101.
- Dweck, C. S. (1999). *Essays in social psychology. Self-theories: Their role in motivation,
 personality, and development*. Ney York, NY, US: Psychology Press.
- Dweck, C. S. (1986). Motivational processes affecting learning. *American Psychologist,*
41(10), 1040– 1048.
- (Elliott, R., Strenta, C., Adair, R., Matier, M., and Scott, J. (1996) “The Role of Ethnicity
 in Choosing and Leaving Science in Highly Selective Institutions.” *Research in
 Higher Education, Vol. 37, No. 6, pp 681-709.*
- Evans, N., Forney, F., & Guido-DiBrito, F. (1998). *Student development in college*. San
 Francisco: Jossey-Bass.

- Farsides, T., & Woodfield, R. (2003). Individual differences and undergraduate academic success: the roles of personality, intelligence and application. *Personality and individual differences*, 40, 1225-1243.
- Freeman, S., Eddy, S. L., McDonough, M., Smith, M., K., Okoroafor, N., Jordt, H., & Wenderoth, M. P. (2014). Active learning increases student performance in science, engineering, and mathematics. *Proceeding of the National Academy of Science of the United States of America* 11(23), 8410-8415.
- Frey, M., & Determan, D. (2004). Scholastic assessment or General cognitive ability? The relationship between scholastic assessment test and general cognitive ability. *Psychological science*, 15, 373-378.
- Fry, B. (2005). Buzz words. The APCA Student Activities Journal.
- Gardner, H. (1983) *Frames of mind*. New York: Basic Book Inc.
- Gardner, J. N., and Jewler, A. J. (1995). *Your College Experience: Strategies for Success* (2nd ed.). Boston: Wadsworth-Thomson Publishing Company.
- Geocaris, C., & Goad, B. (2004). *Report on student involvement in activities and clubs*. Retrieved September 03, 2016, from www.south.hinsdale86.org
- Germain, C. B. (1991). *Human behavior in the social environment: An ecological view*. New York: Columbia University Press.
- Gladieux, L. E., and Swail, W. S. (1998). Financial Aid is Not Enough: Improving the Odds of College Success. *College Board Review*, (185): 16-21, 30-32.
- Gordon, E. W. (1999). *Education and Justice: A View From the Back of the Bus*. New York: Teachers College Press.

- Guest, G., Bunce, A. and Johnson, L. (2006) How Many Interviews Are Enough? An Experiment with Data Saturation and Variability. *Field Methods*, 18, 59-82.
<http://dx.doi.org/10.1177/1525822X05279903>
- Gutierrez, R. (2000). Advancing African American, urban youth in mathematics: Unpacking the success of one math department. *American Journal of Education*, 109(1), 63-111.
- Hackett, G., & Betz, N. E. (1989). An exploration of the mathematics self-efficacy/mathematics performance correspondence. *Journal for Research in Mathematics Education*, 20(3), 261-273.
- Hill, W. (2002). Learning: a survey of psychological interpretations(7 ed.). Boston: Allyn and Bacon., Judge, T., Jackson, C., Shaw, J., Scott, B., & Rich, B. (2007). Self-efficacy and work-related performance: the integral role of individual differences. *Journal of applied psychology*, 92, 107-127.
- Hoffman, N. (2005). *Add and Subtract: Dual Enrollment as a State Strategy to Increase Postsecondary Success for Underrepresented Students*. Boston: Jobs for the Future.
- Holland, A., & Andre, T. (1987). The effects of participation in extracurricular activities in secondary school: What is known, what needs to be known. Iowa State University. p. 117-146. (ED 290 804)
- Hossler, D., and Schmit, J. (1995). The Indiana Postsecondary-Encouragement Experiment. *New Directions for Higher Education*, 89: 27-39.

- Hossler, D., Schmit, J., and Vesper, N. (1999). *Going to College: How Social, Economic, and Educational Factors Influence the Decisions Students Make*. Baltimore: Johns Hopkins University Press.
- House, J. (2000). The effects of student involvement on the development of academic self-concept. *Journal of Social Psychology*, 140, 261-263.
- Hughes, K. L., Karp, M. M., Fermin, B. J., and Bailey, T. R. (2005). *Pathways to College Access and Success*. Office of Vocational and Adult Education. Washington, DC: U.S. Department of Education.
- Husserl, E. (1970). *Trans D Carr Logical investigations*. New York, Humanities Press.
- Hutley, K. (2004). *Alexander Astin's Theory of Involvement: A summary*. Retrieved November 14, 2016, from <http://www.cat.ilstu.edu/conf/astin.html>
- Institute for Higher Education Policy (IHEP). (1995, June). *The Next Step: Student Aid for Student Success*. Washington, DC: Author.
- Institute for Higher Education Policy (IHEP) (2001, February). *Getting Through College: Voices of Low-Income and Minority Students in New England*. Washington, DC: Author.
- Johnson, D., & Renwick, D. (1983). Academic achievement of student athletes. *Athletic Administration*, 18(1), 22-24.
- Johnson, M. L. (1989). Minority differences in mathematics. In M. M. Lindquist (Ed.), *Results from the fourth mathematics assessment of the National Assessment of Educational Progress* (pp. 135-148). Reston, VA: National Council of Teachers of Mathematics.

- Joshi, A., Pan, A., Murakami, M., & Narayanan, S. (2010). Role of computers in educating young children: U.S. and Japanese teachers' perspectives. *Computers in the Schools*, 27(1), 5-19. doi:10.1080/07380560903536249
- Kayatin, W. (2005). *Engaging academe in student life*. The APCA Student Activities Journal.
- King, M. (2006). Finding balance. *Campus Activities Programming*, 38(6).
- Kleiner, B., and Lewis, L. (2005). *Dual Enrollment of High School Students at Postsecondary Institutions: 2002-03*. (NCES 2005-008). U.S. Department of Education. Washington, DC: National Center for Education Statistics.
- Komaraju, M., & Karau, S. (2005). The relationship between the Big Five personality traits and academic motivation . *Personality and individual differences*, 8, 557-567.
- Komaraju, M., Karau, S., & Schmeck, R. (2009). Role of the big five personality traits in predicting college students' academic motivation and achievement. *Learning and individual differences*, 19, 47-52.
- Kuh, G. D., Kinzie, J., Schuh, J. H., and Whitt, E. J. (2005). *Student Success in College: Creating Conditions That Matter*. San Francisco: Jossey-Bass.
- Lambert, E. (2004). How can we better help student success? *The APCA Student Activities Journal*.
- Lau, K. (2003). Institutional Factors Affect Student Retention. *Journal of Experimental Education*. 1(124).

- Littleton, R. A. (2001, November). *African American student persistence at small colleges in Appalachia: A qualitative study*. Paper presented at the annual meeting for the Study of Higher Education, Richmond, VA.
- Livengood, J. (1992). Students' motivational goals and beliefs about effort and ability as they relate to college academic success. *Research in Higher education*, 33, 247-261.
- May, G. S., & Chubin, D. E. (2003). *A retrospective on undergraduate engineering success for underrepresented minority students*. *Journal of Engineering Education*, 92, 27-39
- Martin, D. (2000). *Mathematics success and failure among African American youth: The roles of socio-historical context, community forces, school influence, and individual agency*. Mahwah, NJ: Erlbaum Associates.
- McCombs, B., & Marzano, R. (1990). Putting the self in self-regulated learning: the self as agent in integrating skill and will. *Educational psychologist*, 25, 51-70.
- McGrath, C. (2002). *Under-developed and over-involved: Unpacking the hidden realities in student leadership development*. Retrieved August 03, 2016, from The Vermont Connection.
- McKenzie, K., & Schweitzer, R. (2001). Who succeeds at university? Factors predicting academic performance in first year Australian university students. *Higher Education Research and Development*, 20(1), 21-33.

- Merisotis, J.P., Richardson, E.S., Sullivan, L.W., Pierce, R.C., Kim, M.M., (2006). *The Educational Effectiveness of Historically Black Colleges and Universities (HBCU's)*. The United States Commission on Civil Rights (USCCR)
- Meyers, D., Olsen, R., Seftor, N., Young, J., and Tuttle, C. (2004). *The Impacts of Regular Upward Bound: Results From the Third Follow-Up Data Collection (NCES 2004-13)*. U.S. Department of Education. Washington, DC: National Center for Education Statistics.
- Miller, R., & Brickman, S. (2004). A model of future-oriented motivation and self-regulation. *Educational Psychology Review*, 16, 9-33.
- Monroe Community College. (2004). *Campus center mission*. Retrieved June 14, 2016, from <http://www.monroecc.edu/depts/stucenter/mission.htm>.
- Moody, V. R. (2004). Sociocultural orientations and the mathematical success of African American students. *Journal of Educational Research*, 97(3), 135-148.
- Moore, J. (2000). Note: Are State-Supported Historically Black Colleges and Universities Justifiable after Fordice? *A Higher Education Dilemma*, 27 Fla. St. U. L. Rev. 547, 550 (2000).
- Muraskin, L. (1997). *"Best Practices" in Student Support Services: A Study of Five Exemplary Sites*. Washington, DC: U.S. Department of Education, Planning, and Evaluation Service.
- Nason-Saltonstall, M. (2004). *Involvement equals retention*. Retrieved November 27, 2016, from NAU SS Student Affairs Datamart.

- National Center for Education Statistics (NCES). *The Condition of Education 2016*. Washington, D.C. National Center for Education Statistics, 2016.
- National Research Council. (1999). *Equity & Adequacy in Education Finance: Issues and Perspectives*. Washington, DC: National Research Council Committee on Education Finance.
- Nora, A., & Cabrera, A. F. (1996). The role of perceptions of prejudice and discrimination on the adjustment of minority students to college. *Journal of Higher Education*, 67(2), 119-148.
- Oakes, J. (1990). Opportunities, achievement, and choice: Women and minority students in science and mathematics. *Review of Research in Education*, 16(1), 153-222.
- Padilla, R. V. (1999-2000). College student retention: Focus on success. *Journal of College Student Retention*, 1, 131-145.
- Papanikolaou, Z. (2003). The freshman experience: High Stress low grades. *Athletic Insight*, 5(4).
- Pascarella, E. T., and Terenzini, P. T. (1991). *How College Affects Students: Findings and Insights From Twenty Years of Research (1st ed.)*. San Francisco: Jossey Bass Publishers.
- Pascarella, E. T., & Terenzini, P. T. (1978). Student-faculty informal relationships and freshman year educational outcomes. *Journal of Educational Research*, 71(4), 183-189.

- Pascarella, E. T., & Terenzini, P. T. (2005). *How college affects students: A third decade of research* (Vol. 2). San Francisco: Jossey-Bass.
- Pathways to College Network. (2004). *A Shared Agenda: A Leadership Challenge to Improve College Access and Success*. Boston: The Education Resources Institute.
- Patton, M. Q. (2002). *Qualitative research and evaluation methods* (3rd ed). Thousand Oaks, CA: Sage Publications.
- Pennington, H. (2004, December). *Fast Track to College: Increasing Postsecondary Success for All Students*. Boston: Jobs for the Future.
- Powell, L. (1990). Factors Associated with the under representation of African Americans in mathematics and science. *Journal of Negro Education*, 59(3), pp 292 – 298.
- Powell-Mikle, A. (2001, April). *African American mathematic student perceptions of peers, education, and mathematical experiences*. Paper presented at the annual meeting of the American Educational Research Association, Seattle, WA.
- Pritchard, M., & Wilson, G. (2003). Using emotional and social factors to predict student success. *Journal of college student development*, 44, 18-28.
- Ravitch, D. (2003). *Brookings papers on education policy*. Retrieved October 04, 2016, from www.brook.edu
- Redd, K. (2000)“HBCU Graduates: Employment, Earnings and Success after College.” *USA Group Foundation New Agenda Series*, 2000, 2(4).
[<http://www.luminafoundation.org/publications/hbcu.pdf>].

- Reynolds, A. J., and Walberg, H. J. (1992). A Process Model of Mathematics Achievement and Attitude. *Journal for Research in Mathematics Education*, 23(4): 306-328.
- Ridgell, S., & Lounsbury, J. (2004). Predicting academic success: general intelligence, "big five" personality traits and work drive. *College student journal*, 38, 607-618.
- Rizer, M. (2005). When students are parents. *The Chronicle of Higher Education*, 52(17).
- Roebuck, J. B., and K. S. Murty (1993). *Historically Black Colleges and Universities*, Praeger Publisher.
- Rogers Elliott, A. Christopher Strenta, Russell Adair, Michael Matier, and Jannah Scott, (2006) "The Role of Ethnicity in Choosing and Leaving Science in Highly Selective Institutions". *Research in Higher Education*, vol. 37, No. 6, pp. 684, 700.
- Rogoff, B. (1995). Observing sociocultural activity in three planes: Participatory appropriation, guided participation, and apprenticeship.
- Wertsch, P. del Rio, & A. Alvarez (Eds.), *Sociocultural studies of the mind* (pp. 139-164). Cambridge, UK: Cambridge University Press.
- Roland G. F, Jr., and Greenstone, M. (2007). *The Causes and Consequences of Attending Historically Black Colleges and Universities*. National Bureau of Economic Research Working Paper No. 13036, April 2007, p.1.
- Roman, M. (2007). *Community college admission and student retention*. *Journal of College Admission*. Retrieved August 11, 2016, from <http://www.nacacnet.org>
- Schunk, D. (2004). *Learning theories: an educational perspective (4 ed.)*. Columbus, OH: Merrill/Prentice-Hall.

- Schwandt, T. A. (2000). Three epistemological stances for qualitative inquiry: Interpretivism, hermenutics, and social construction.
- N. K. Denzin & Y. S. Lincoln, (Eds). *Handbook of qualitative research*, p. 189- 213. Thousand Oaks, CA: Sage.
- St. John, E. P. (2003). *Refinancing the College Dream: Access, Equal Opportunity, and Justice for Taxpayers*. Baltimore: Johns Hopkins University Press.
- Strutchens, M. E. (1993). An exploratory study of the societal and ethnic factors affecting sixth-grade African American students' performance in a mathematics class (Doctoral dissertation, University of Georgia, 1993). *Dissertation Abstracts International*, 54, 2084.
- Summak, M. & Samancioglu, M. (2011). *Assessment of technology integration in 112 teachers*. (Doctoral dissertation). Retrieved from ProQuest Dissertations and Theses database. (AAT 3238961)
- Swail, W. S., Cabrera, A. F., Lee, C., and Williams, A. (2005). *Latino Students and the Educational Pipelines: A Three-Part Series. Part III: Pathways to the Bachelor's Degree for Latino Students*. Stafford, VA: Education Policy Institute.
- Terenzini, P. T., Springer, L., Yaeger, P., Pascarella, E. T., and Nora, A. (1996). First-Generation College Students: Characteristics, Experiences, and Cognitive Development. *Research in Higher Education*, 37(1): 1-22.
- Terenzini, P. T., & Pascarella, E. T. (1980). Student/faculty relationships and freshman year educational outcomes: A further investigation. *Journal of College Student Personnel*, 21(6), 521-528.

- Thomas, S. (2005). Engaging students with school life. *Youth Studies Australia*, 24(1).
- Thompson, L. R., & Lewis, B. F. (2005). Shooting for the stars: A case study of the mathematics achievement and career attainment of an African American male high school student. *The High School Journal*, 88(4), 6-18.
- Tierney, W. G., and Hagedorn, L. S. (2002). *Increasing Access to College: Extending Possibilities for All Students*. Albany, NY: State University of New York Press.
- Tierney, W. G., Corwin, Z. B., and Colyar, J. E. (Eds.) (2004). *Preparing for College: Nine Elements for Effective Outreach*. Albany, NY: State University of New York Press.
- Tinto, V. (1987). *Leaving College: Rethinking the Causes and Cures of Student Attrition*. Chicago: University of Chicago Press.
- Tinto, V. (1998). Colleges as communities: Taking research on student persistence seriously. *Review of Higher Education*, 15(2), 109-127.
- Tisdell, E. J. (2000). *Feminist pedagogies*. In E. Hayes, D. Flannery, & Associates (Eds.), *Women as learners* (pp. 155-184). San Francisco: Jossey-Bass.
- Trevas, D. (1996). *Getting involved key to getting through college*. Retrieved October 5, 2016, from <https://www.tntech.edu>
- Truett, P. (2006). Factors impacting technology implementation of math Learning and Development. *Journal of Math*, 3(1), 121. doi:10.5296/ijld.v3i1.3154
- Tuckman, B. (2003). The effect of learning and motivation strategies training on college students' achievement. *Journal of college student development*, 44, 430-437.

- Upcraft, M., Gardner, J., & Barefoot, B. (2005). *Challenging and supporting the first-year student: A handbook for improving the first year of college*. San Francisco: John Wiley & Sons, Inc.
- United States Commission on Civil Rights (2006) *Performance and Accountability*
Publisher: U.S. Commission on Civil Rights. Washington, DC
- Venezia, A., Kirst, M. W., and Antonio, A. L. (2003, March). *Betraying the College Dream: How Disconnected K–12 and Postsecondary Education Systems Undermine Student Aspirations*.
- Volkwein, J. F., Valle, S., Parmley, K., Blose, G., and Zhou, Y. (2000, May). *A Multi-Campus Study of Academic Performance and Cognitive Growth Among Native Freshman, Two-Year Transfers, and Four-Year Transfers*. Paper presented at the annual meeting of the Association for Institutional Research, Cincinnati, OH.
- Von Glasserfeld, E. V. (1995). *Radical constructivism: A way of knowing and learning*. *Studies in mathematics education series: 6*. London: The Falmer Press.
- Vygotsky, L. S. (1994). *Thought and language*. Cambridge, MA: M.I.T. Press.
- Vygotsky, L. S. (1981). *The genesis of higher mental functions*. J. V. Wertsch (Ed.), *The concept of mental activity in Soviet psychology* (pp. 144-188). Armonk, NY: M. E. Sharpe.
- Watkins, T. J. (1997). Teacher Communications, Child Achievement, and Parent Traits in Parent Involvement Models. *Journal of Educational Research*, 91(1): 3-14.

- Weiner, B. (1974). *Attribution theory, achievement motivation, and the educational process*. Review of educational research, 42(2), 203-215
- Weiner, B. (1979). A theory of motivation for some classroom experiences. *Journal of Educational Psychology*, 71(1), 3-25.
- Weller, L. D. (1984). *Attitude toward grade inflation: A survey of private and public colleges of education*.
- Wenglinsky, H. (1998). Finance Equalization and Within-School Equity: The Relationship between Education Spending and the Social Distribution of Achievement. *Educational Evaluation and Policy Analysis*, 20(4): 269-283.
- Wolfe, R., & Johnson, S. (1995). Personality as a predictor of college performance. Educational and psychological styles in addition to personality traits. *Personality and educational differences*, 33, 445-458.
- Zhang, A., & Aasheim, C. L. (2011). Academic Success Factors: An IT Student Perspective. *Journal of Information Technology Education* 10, 309-330.
- Zhang, Z., & RiCharde, R. (1997). *Learning-thinking style inventory: LISREL and multivariate analyses*. Paper presented at the Annual Meeting of the American Educational Research Association. (ERIC Document Reproduction Service No. ED410264)

APPENDICES

APPENDIX A: Additional factors related to student Success

Table 1: Additional factors related to student success

Factors	References
Cognitive learning styles: concrete sequential, abstract sequential, abstract random, and concrete random.	Ross et al., 2001
Learning styles: perceptual modality, distractibility, meta- cognition, analytic global tendency.	Zhang & RiCharde, 1997
Ability, effort, task, luck	Boekaerts, Otten, & Voeten, 2003
Ability, effort, task, luck, culture, gender, course subject	Yan & Gaier, 1994
Reading and writing abilities, task difficulty, study habits	Ritchey & Lewis, 1986
Effort, self-efficacy, loneliness, coherence, mood, hope	Lackaye & Margalit, 2006
Goal orientation, self-efficacy and self-regulation	Niemczyk & Savenye, 2001
Self esteem	Feick & Rhodewalt, 1997
Anxiety, locus control, critical thinking	Hulse et al., 2007
Academic, psychosocial, cognitive, demographic factors	McKenzie & Schweitzer, 2001
School expenditure, teacher ability, teacher education, teacher experience, teacher salary, teacher/student ratio, school size	Greenwald, Hedges, & Laine, 1996
Class size	Ehrenberg, Brewer, Gamoran, & Willms, 2001
Age, gender, type of high school attended, nationality, parental education, number of siblings, siblings' education, living arrangement changes	Demeulemeester & Rochat, 1995
Parental education and employment, student's procrastination of task, value of task, anxiety, classroom climate, support from peers, task load, instruction quality	Bruinsma & Jansen, 2007
Family income, parental education, parental involvement, and hostility	Melby & Conger, 1996
Parental practices in support, reasoning, punitiveness, monitoring, and autonomy granting	Henry, Martinko, & Pierce, 1993

Three types of orientation (self, interaction, task), four cognitive styles (concrete experience, reflective observation, abstract conceptualization, active experimentation), attitude to- ward course and computer use; philosophy in (vocation, academic, social interaction, individual identity)	Kevin & Liberty, 1975
Support networks, depression, life events	Blumberg, 1984
Field of study, quantity of course work, and GPA of teacher	Denton et al., 1981
Amount of time spent, teacher characteristics, teacher/student interaction, classroom control, instructional (organization, integration, materials, articulation)	Connolly & McGrail, 1978
Teachers' behavior, teaching methods, commitment to study, learning environment, content of study, psychological problems, relationship with family, concern with field of study and future career, time management.	Aysan et al., 1996
Relationship with teacher, interests in assignments, feeling of competence	Daniels & Arapostathis, 2005
Student-teacher relation and interaction	McGregor, 2007
Perceived faculty support	Shelton, 2003
Teacher salaries, school spending, reduced-price or free lunch	Costrell, Hanushek, & Loeb, 2008
State program: Families and Communities Equals Success in North Carolina	Fernandez, Campbell, & Hon- ness, 2000
Retention policy	Bowman, 2005
Social and economic factors	Matuszek & Haskin, 1978 Gradstein & Justman, 2000 Grossman, 2008
Social cohesion	Mixon & Treviño, 2005
Societal health including human capital, knowledge capital, and health capital	Mathews & Hackett, 1997
Success of athletic programs	Howard, Borland, Johnson, & Baker, 2001
Government policy: education accountability in Alabama	Wince & Borden, 1995
Suspending policy	
Social climate, academic facilities, co-curricular activities, advising, getting into classes (registration, financial aid, course availability, textbook)	Volkwein, Valle, Parmley, Blose, & Zhou, 2000
Institutional factors, academic integration, social integration, campus climate, financial status	

Accountability policy	Hanushek& Raymond, 2005
Study time influences academic performance but is moderated by study habits. Planning/scheduling not as important for short-term success.	Nonis& Hudson, 2010
Tracking policy to place and group students based on ability	Hanushek&Wobmann, 2006

APPENDIX B: Permission to use Semi-Structured Interview Guide

Re: Permission to use the Semi-Structured Interview Guide.

You replied on Tue 10/2/2018 at 4:59 PM

grolland<grolland@comcast.net>
Mon 10/1/2018 7:30 PM

You have my permission to make use of my questioning guide. Much success to you with your study.

Gertrude Rolland Cuthbert

Sent from my Verizon, Samsung Galaxy smartphone

----- Original message -----

From: Davon Williams <wdavon@hotmail.com>

Date: 10/1/18 2:56 PM (GMT-05:00)

To: grolland@comcast.net

Subject: Permission to use the Semi-Structured Interview Guide.

Hello Dr. Rolland, I hope this e-mail finds you doing well.

My name is Davon Williams and I am a Doctoral Student at Tennessee State University in the Educational Administration Department.

I'm writing to request permission to use your interview guide with modifications for my Dissertation study titled: Perceptions of African American Students about their Success in Mathematics at a Historically Black College and University (HBCU).

Being that my target population consists of African American STEM College students, some questions may be asked differently and some may be eliminated to gather the necessary feedback from the study participants.

In addition, credit will be given to you as the creator of this questioning guide.

Please let me know your decision on this matter.

I can be reached at (615) 424-6796 or by e-mail if you have any questions.

Thank You.

Davon

APPENDIX C: Interview Guide

Interview Guide

1. Tell me about yourself
2. What is your perception of mathematical success and how do you define mathematical success?
3. Describe what it means to be confident in your ability to successfully pass mathematics classes?
4. In what ways will being academically successful in mathematics affect your progress (in your major & towards graduation)
5. Discuss strategies you have used in helping you to become successful in mathematics classes?
6. Explain the contribution your parents may have had regarding your association with mathematics
7. In addition to strategies, you have implemented what other strategies would you advise for success in mathematics and why?
8. How do you feel about traditional class settings vs. non-tradition class settings (Hybrid courses/technology base vs non-hybrid)?
9. In what ways does your collegiate involvement contribute or distract from the success of African American Students in Mathematics? (Sports, Band, Clubs, and Organization, etc)

10. What do you feel is necessary for college professors to do in order to help African American Students attain success in mathematics?
11. What may prevent female African American STEM Students from having success in freshmen level mathematics courses?
12. Describe to what extent has high school has prepared or not prepared you to succeed in freshmen college-level mathematics.
13. Describe how technology was used in your high school mathematics experience and what role does it provide in your collegiate mathematics experience?
14. Is there something you thought was necessary for me to address but I did not and would you like to add any additional information on the subject regarding mathematical success

APPENDIX D: Demographic Questionnaire
Demographic Questionnaire

Below is a questionnaire that provides me with some information about who you are and your background. The purpose of this questionnaire is twofold, a) to acknowledge your interest in a dissertation study about African American STEM student success in Mathematics and 2) determine if you qualify to participate in the study. Information collected and used during this study will be kept confidential and nothing will be divulged without your written permission and consent. For the purpose of this study and to allow you to remain anonymous I have provided you with an option to use a pseudonym of your choice to identify you and your contribution to this study

1. What is your name? _____

1a. What is your classification? _____

1b. What is your e-mail address? _____

1c. What is your telephone contact? _____

2. Please list a Pseudonym you would like to use. _____

3. What is your gender?

() Female

() Male

() Other

4. What is your current Major? _____

5. When do you anticipate you will graduate? _____

6. Please list all math classes taken in high school and grade received.

Class: _____ Grade: _____

Class: _____ Grade: _____

Class: _____ Grade: _____

Class: _____ Grade: _____

7. Please list math courses you have taken and/or are taking now and provide your grade received or grade you expect to receive.

Class: _____ Grade: _____

Class: _____ Grade: _____

Class: _____ Grade: _____

8. Does either one of your parents have a college degree

Yes () or No () No comment ()

9. Did you receive Financial Aid?

Yes () or No () No comment ()

10. If chosen to participate in this study and answer a series of open-ended questions about your math perceptions and experiences would you be willing to participate in the interview? Yes () or No ()

APPENDIX E: Department Head Instructor Request

December 3, 2019

Dr. Nolan McMurry, Jr., Professor
Department Head, Mathematics Department
Tennessee State University
3500 John A Merritt Blvd
Nashville, TN 37209

Dear Dr. McMurry:

I am working on a dissertation titled, Perceptions of African American students about their success in Mathematics at a Historically Black College and University (HBCU) and I am seeking permission to speak with some of your mathematics classes in regard to participation in this study.

The primary purpose of this study is to gather and investigate the perceptions of African American STEM students regarding what constitutes success in mathematics, and identify how factors associated with mathematical success has contributed to their individual success in mathematics.

By listening to the voices of these students both the university and the mathematics department could benefit by adjusting strategies, programs, and procedures that can address a student shortage in the mathematics area and in the broader sense the STEM area. In addition, teaching strategies can be employed to fully address the need of African American students who seldom choose a college major relating to STEM and ensure that the majority and not the minority of mathematics students are successfully passing their classes.

Thanks in advance for your consideration regarding this communication. If there are any additional questions or concerns please feel free to contact me at e-mail: dwilliams02@tnstate.edu, or phone: (615) 424-6796.

Sincerely,

Davon W. Williams

Doctoral Student
College of Education, Tennessee State University

cc: Dr. Kirmanj Gundi, Dissertation Chair, TSU

Approved by:

Print your name and title

Signature

Date

APPENDIX F: Department Head and Instructors Approvals

December 16, 2019

Dr. Nolan McMurry, Jr., Professor
Department Head, Mathematics Department
Tennessee State University
3500 John A Merritt Blvd
Nashville, TN 37209

Dear Dr. McMurry:

I am working on a dissertation titled, Perceptions of African American Students about their success in Mathematics at a Historically Black College and University (HBCU) and I am seeking permission to speak with some of your mathematics classes in regard to participation in this study.

The primary purpose of this study is to gather and investigate the perceptions of African American STEM students regarding what constitutes success in mathematics, and identify how factors associated with mathematical success has contributed to their individual success in mathematics.

By listening to the voices of these students both the university and the mathematics department could benefit by adjusting strategies, programs and procedures that can address a student shortage in the mathematics area and in the broader sense the STEM area. In addition teaching strategies can be employed to fully address the need of African American students who seldom choose a college major relating to STEM and ensure that the majority and not the minority of mathematics students are successfully passing their classes.

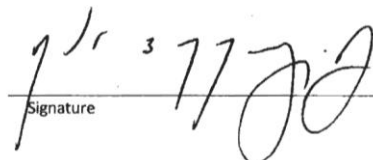
Thanks in advance for your consideration regarding this communication. If there are any additional questions or concerns please feel free to contact me at e-mail: dwilliams02@tnstate.edu, or phone: (615) 424-6796.

Sincerely,

Davon W. Williams
Doctoral Student
College of Education, Tennessee State University

cc: Dr. Kirmanj Gundi, Dissertation Chair, TSU

Approved by:


Signature

1/17/2020
Date

December, 16, 2019

Dr. Kothandaraman Ganesan, Associate Professor
Tennessee State University
3500 John A Merritt Blvd
Nashville, TN 37209

Dear Dr. Ganesan:

I am working on a dissertation titled, Perceptions of African American Students about their success in Mathematics at a Historically Black College and University (HBCU) and I am seeking permission to speak with some of your mathematics classes in regard to participation in this study.

The primary purpose of this study is to gather and investigate the perceptions of African American Mathematics students regarding what constitutes success in mathematics, and identify how factors associated with mathematical success has contributed to their individual success in mathematics.

By listening to the voices of these students both the university and the mathematics department could benefit by adjusting strategies, programs and procedures that can address a student shortage in the mathematics area and in the broader sense the STEM area. In addition teaching strategies can be employed to fully address the need of African American students who seldom choose a college major relating to STEM and ensure that the majority and not the minority of mathematics students are successfully passing their classes.

Thanks in advance for your consideration regarding this communication. If there are any additional questions or concerns please feel free to contact me at e-mail: dwilliams02@tnstate.edu, or phone: (615) 424-6796.

Sincerely,

Davon W. Williams
Doctoral Student
College of Education, Tennessee State University

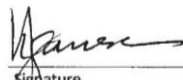
cc: Dr. Kirmanj Gundi, Dissertation Chair, TSU

cc: Dr. Kirmanj Gundi, Dissertation Chair, TSU

Approved by:

K. GANESAN

Print your name and title


Signature

12/16/19
Date

January 9, 2020

Dr. Wanda Payne, Assistant Professor
Tennessee State University
3500 John A Merritt Blvd
Nashville, TN 37209

Dear Dr. Payne:

I am working on a dissertation titled, Perceptions of African American Students about their success in Mathematics at a Historically Black College and University (HBCU) and I am seeking permission to speak with some of your mathematics classes in regard to participation in this study.

The primary purpose of this study is to gather and investigate the perceptions of African American Mathematics students regarding what constitutes success in mathematics, and identify how factors associated with mathematical success has contributed to their individual success in mathematics.

By listening to the voices of these students both the university and the mathematics department could benefit by adjusting strategies, programs and procedures that can address a student shortage in the mathematics area and in the broader sense the STEM area. In addition teaching strategies can be employed to fully address the need of African American students who seldom choose a college major relating to STEM and ensure that the majority and not the minority of mathematics students are successfully passing their classes.

Thanks in advance for your consideration regarding this communication. If there are any additional questions or concerns please feel free to contact me at e-mail: dwilliams02@tnstate.edu, or phone: (615) 424-6796.

Sincerely,

Davon W. Williams
Doctoral Student
College of Education, Tennessee State University

cc: Dr. Kirmanj Gundi, Dissertation Chair, TSU

Approved by:

Wanda Payne
Print your name and title

Wanda Payne
Signature

1/9/20
Date

APPENDIX G: IRB Approval Letter and Amendment



"Think. Work. Serve."

Research and Sponsored Programs
3500 John A. Merritt Boulevard
Nashville, Tennessee 37209-1561
Office: (615) 963-7631
Fax: (615) 963-5068

To: Davon Williams

Department: Educational Leadership

Re: Protocol # HS-2019-4393

Date: Wednesday, January 22, 2020

From: Monique McCallister, Ph.D., Institutional Review Board

The document listed below has been carefully reviewed and found to be compliant with OPRR document title 45, Code of Federal Regulations part 46, the protection of human subjects, as amended by Federal policy, effective August 19, 1991. This project is **approved** as it presents minimal to no research risks to human participants involved. Please make note, that any deviations in the administration of the protocol, accidental or otherwise should be reported to the IRB as soon as possible. The FWA for Tennessee State University is #FWA00007692, which is effective from July 15, 2016 to September 18, 2022.

Perceptions of African American Students about their Success in Mathematics at a Historically Black College and University (HBCU) (HS-2019-4393)

This approval is valid for one year from the date indicated above. Continuation of research beyond that date requires re-approval by the Institutional Review Board.

Please contact me at 615-963-7619 or e-mail irb@tnstate.edu for additional information.

Monique McCallister, Ph.D.
mmccalli@tnstat.edu



"Think. Work. Serve."

Research and Sponsored Programs
3500 John A. Merritt Boulevard
Nashville, Tennessee 37209-1561
Office: (615) 963-7631
Fax: (615) 963-5068

To: Davon Williams

Department: Educational Leadership

Re: Protocol #HS-2019-4393 Amendment

From: Monique McCallister, Ph.D., Institutional Review Board

Date: Wednesday, February 05, 2020

The document listed below has been carefully reviewed and found to be in compliance with OPRR document title 45, Code of Federal Regulations part 46, the protection of human subjects, as amended by Federal policy, effective August 19, 1991. This project is **approved** as it presents minimal or no research risks to the pool of impending human subjects. Please make note, that any deviations in the administration of the protocol, accidental or otherwise should be reported to the IRB as soon as possible. The FWA for Tennessee State University is #FWA00007692, which is effective from July 15, 2016 to September 18, 2022.

Perceptions of African American Students about their Success in Mathematics at a Historically Black College and University (HBCU) (HS-2019-4393)

This approval is valid for one year from the date indicated above. Continuation of research beyond that date requires re-approval by the Institutional Review Board.

Please contact me at 615-963-7619 or e-mail irb@tnstate.edu for additional information.

Monique McCallister, Ph.D.
mmccalli@tnstat.edu

APPENDIX H: Informed consent statement

INFORMED CONSENT STATEMENT: Perceptions of African American Students about their Success in Mathematics at a Historically Black College and University (HCU)

INTRODUCTION

The College of Education supports the practice of protection for human subjects participating in research. The following information is provided to help you decide whether you wish to participate in the present study. You retain the right to refuse to sign this form and not participate in this study. You should be aware that even if you consent to participate in this study, you may withdraw from this study at any time without consequence. If you choose to withdraw from this study, it will not affect your relationship with this department, your school/program, or professor/instructor.

PURPOSE

The purpose of the study is to explore the perceptions of African American STEM students as it relates to success in Mathematics in a university setting. In addition, noted factors will be addressed to identify their importance to your mathematical success. Your participation in this study is voluntary and not required for any portion of your course grade or program completion. There is no extra credit, nor benefit to your grade in the class.

PROCEDURES

During the initial class visit, you will be provided with information about the study, and if interested be required to fill out a demographic questionnaire. Filled out demographic questionnaires will be collected number stamped and processed through a random selection tool using excel. Selected participants will be notified by e-mail and/or phone to set up a time and place to conduct the interview sessions. After data is analyzed a follow-up session will take place with each selected participant to ensure accuracy in the interpretation of the interview and correct where necessary. The total time for participation will be between 30-90 minutes over the course of the entire semester.

RISKS

There are no anticipated risks in completing this survey.

BENEFITS & COMPENSATION

There are no individual benefits for survey participants. There is no financial compensation for students.

PARTICIPANT CONFIDENTIALITY

Student participants will be contacted directly by the researcher. Data collected will be done so anonymously. Any information collected will be kept secured in a locked file and destroyed after a period of 2 years.

REFUSAL TO SIGN CONSENT

You are not required to sign this Consent form and you may refuse to do so without affecting your right to participate in any programs or events at **Tennessee State University** or any services you are receiving.

CANCELLING THIS CONSENT

You may withdraw your consent to participate in this study at any time. If you choose to withdraw from the study before data collection is completed, any collected data will be destroyed and not used. You may contact the researcher, Mr. Davon Williams, for further questions or to withdraw from the study at dwilliams02@tnstate.edu.

CONSENT: Completing the consent form is required to further participate in the study.

By my signature, I affirm that I am at least 18 years old and a student at Tennessee State University.

Print Participant's Name Date

Participant's Signature Date

Email: _____

APPENDIX I: CITI Completion Certificate

		Completion Date 14-Dec-2017 Expiration Date 13-Dec-2020 Record ID 25534888
This is to certify that:		
Davon Williams		
Has completed the following CITI Program course:		
Social & Behavioral Research - Basic/Refresher (Curriculum Group) Social & Behavioral Research (Course Learner Group) 1 - Basic Course (Stage)		
Under requirements set by:		
Tennessee State University		
 Collaborative Institutional Training Initiative		
Verify at www.citiprogram.org/verify/?w304e8e7d-0095-4b35-84f1-8d920d8b0a4-25534888		

VITA

Davon W. Williams

177 Star Blvd
Madison, TN 37115
Tel: (615) 424-6796
E-mail: wdavon@hotmail.com

OBJECTIVES: To obtain a Math Education teaching position or a research position in the field of Math Education that will allow me to utilize and expand acquired knowledge.

EDUCATION:

November 2020 Doctor of Education (Ed.D), Education (Education Leadership)
Tennessee State University, Nashville, TN.

May 2011 Education Specialists (Ed.S), Education (Administration & Supervision)
Tennessee State University, Nashville, TN.

Aug 2005 Master of Science, Mathematical Sciences
Tennessee State University, Nashville, TN.

Aug 2001 Bachelor of Science, Physics.
Tennessee State University, Nashville, TN.

May 1997 Associate of Arts, Pre-Engineering. (Electrical Concentration)
College of the Bahamas, Nassau, Bahamas.

Major Courses Education Specialists: Leadership/Interpersonal Relationships, Statistics for Education Administration, Theory/Principles of Education Administration, Professional Negotiations, Legal Problems, Financial Management in Educational Administration.

Mathematics: Linear Algebra I & II, Advance Calculus I & II, Advance Differential Equations I, Probability & Statistics I, Complex Analysis I, Modern Algebra, Number Theory.

Physics: General Physics I, II & III, Thermodynamics, Electromagnetic Theory I & II, Optics, Quantum Mechanics I & II, Modern Physics I.

Engineering: Digital Design, Engineering Graphics, Statics, Dynamics,

Circuits I & II, Transport Phenomena, Introduction to Design, Material Science, Technical Writing.

Classes Instructed: Contemporary Math, College Algebra I & II, Intermediate Algebra Pre-Calculus I & II, Calculus with Applications, Calculus for Business, Business Mathematics, Statistics, and Trigonometry.

WORK EXPERIENCE:

Aug-Dec **Mathematical Instructor,** Physics & Math Department.
2015, 2016, Tennessee State University, Nashville, TN.
2017, 2018, 2019

- Instructed first-year Algebra, Pre-Calculus, and Contemporary Math classes.
- Maintained accurate records of students' progress and attendance.

Aug-Dec **Mathematical Instructor,**
2014 John A. Gupton College, Nashville, TN.

- Instructed Introductory Business Math classes.
- Maintained accurate records of students' progress and attendance.

Aug –May **Graduate Assistant,** Academic Success Program (SAFRA Grant)
2013-2015 Tennessee State University, Nashville, TN.

- Provided tutorial sessions in Mathematics and Physics classes
- Provide help with data analysis of student progress for reports
- Assist students with beneficial study habits for academic success and provided resources to help students grasp content knowledge in course work

WORK EXPERIENCE:

Aug -May **Unit Academic Advisor,** TRIO Programs
2005-2013 Tennessee State University, Nashville, TN.

- Receive and maintain accurate record of students for APR.
- Provided academic and career counseling.
- Provided tutorial/teaching in Math and Physics classes
- Assisted in preparing students for ACT's, SAT's EOC's, and TCAP.
- Provided exposure to college campuses and academic programs.

- Aug-May**
2011-2013 **Mathematical Instructor**, Physics & Math Department.
Tennessee State University, Nashville, TN.
- Instructed first-year Algebra, Pre-Calculus, and Contemporary Math classes.
 - Maintained accurate records of students' progress and attendance.
- Jan-August**
2006-2006 **Mathematical & Science Teacher**
Nashville Christian Advancement Academy, Nashville, TN.
- Taught courses in Mathematics and Science.
 - Maintained accurate records of students' progress.
 - Provided mentoring to students.
 - Provided ACT Preparation to High School Students.
- Aug-Dec**
Department.
2003-2005 **Mathematical & Science Graduate Associate**, Physics & Math
Tennessee State University, Nashville, TN.
- Instructed first-year Algebra and Calculus level courses.
 - Maintained accurate records of student's progress.
- Aug-Aug**
Department.
2002-2003 **Mathematical & Science Tutorial Associate**, Physics & Math
Tennessee State University, Nashville, TN.
- Provided tutorial services for Physics and Math courses.
 - Substitute taught first and second-year math courses.
- Aug-Aug**
2002-2003 **Student Support Counselor**, TRIO Programs – Educational Talent
Search
Tennessee State University, Nashville, TN.
- Assisted students with academic support counseling.
 - Taught Physics and Math courses
 - Provided personal counseling.
- May-Aug**
Summer 2002 **Student Support Counselor**, TRIO Programs – Upward Bound
Tennessee State University, Nashville, TN.
- Assisted students with academic and peer counseling.
 - Mentored Students and Taught Physics and Math courses
 - Resident Counselor.
- Sep-Dec**
2001- 2001 **Assembly Line /Product Build Operator.**
Dell Computers, Lebanon, TN
- Assembled Computers on the line in production.

May-Aug **Research Assistant**, Center of Excellence in Information Systems.
 Summer 2001 Tennessee State University, Nashville, TN.

- Enhanced research skills and designed web pages.
- Provided weekly reports and flow graphs.
- Setup up concept maps.

May-Aug **Research Assistant**, College of Engineering & Technology.
 Summer 2000 Tennessee State University, Nashville, TN.

- Designed and built webpages.
- Installed and upgraded computer systems.
- Setup computer networking capabilities.

COMPUTER SKILLS: Windows 7, XP, 2000, NT, 98 OS, Microsoft Office 2010, XP, 2000, 98, Word, Excel, PowerPoint & FrontPage, HTML, ColdFusion, Dreamweaver 3 & 4, SPSS 18 & 19, Contribute, Sony Vegas Pro, DVD Architect, Some C-Programming & JAVA.

ACTIVITIES: N.S.B.P., Tennessee Academy of Science, Physics & Math Club.
 Member of Tennessee Association of Special Programs (TASP) and
 Southeastern Association of Educational Opportunity Program Personnel
 (SAEOPP).

AWARDS: B.E.C. Scholarship Recipient 1993, Graduate Assistant Scholarship
 2003, 2013 Student Support Counselor Award 2003, 2014, Alpha Phi Alpha
 Outstanding Award 1993.

PUBLICATIONS: “Travelling Waves of the Perturbed Nonlinear Schrodinger’s
 Equation” International Journal of Differential Equations and Applications
 Volume 9, No. 4, 2004, pages 342-350

PRESENTATIONS: “The Use of X-ray Optics to determine a structural changes of
 LiquidCrystals” Tennessee State University 24th Annual Research Symposium, March
 2002

DISSERTATION TOPIC: Perceptions of African American Students about their
 Success in Mathematics at a Historically Black College and
 University (HBCU).

ProQuest Number: 28153921

INFORMATION TO ALL USERS

The quality and completeness of this reproduction is dependent on the quality and completeness of the copy made available to ProQuest.



Distributed by ProQuest LLC (2021).

Copyright of the Dissertation is held by the Author unless otherwise noted.

This work may be used in accordance with the terms of the Creative Commons license or other rights statement, as indicated in the copyright statement or in the metadata associated with this work. Unless otherwise specified in the copyright statement or the metadata, all rights are reserved by the copyright holder.

This work is protected against unauthorized copying under Title 17,
United States Code and other applicable copyright laws.

Microform Edition where available © ProQuest LLC. No reproduction or digitization of the Microform Edition is authorized without permission of ProQuest LLC.

ProQuest LLC
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 - 1346 USA